

ABSTRACT

Direct Sequence Spread Spectrum (DSSS) is a versatile communication technique with applications in various fields. This project explores the fundamentals of DSSS by building a demonstration board operating at 2 MHz. The board successfully transmits and receives different signals (square waves, Morse code), demonstrating DSSS suitability for HF communication.

The project investigates key DSSS concepts such as spreading, despreading, Pseudo-Noise (PN) sequence generation using linear-feedback shift registers (LFSR), and spectral characteristics. It also analyzes modulation techniques and the factors that influence process gain in DSSS systems.

This project provides a hands-on understanding of spread-spectrum principles. It serves as a valuable educational tool for courses related to communication systems and offers a foundation for further experimentation with DSSS techniques.

This project serves as a valuable educational tool for understanding the fundamental building blocks of spread spectrum communication. It bridges theory and practical implementation, making it particularly relevant for courses like Wired Communication Systems (EE516) and Wireless Communication Systems (EE526). The system offers a foundation for understanding and experimenting with spread-spectrum techniques, applicable to both amateur radio exploration and academic study.

ACKNOWLEDGMENTS

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1. INTRODUCTION

Spread spectrum communication has garnered increasing interest due to its resilience against interference and security advantages. This project aims to explore the design, simulation, and implementation of a Direct Sequence Spread Spectrum (DS-SS) communication system for amateur radio applications. This section gives a scope description and overview of everything included in this Project Report. Also, the purpose for this document is described and system overview along with goal and vision are listed.

1.1. Objectives:

The primary objectives of this mini project are as follows:

- Design a DS-SS generator circuit capable of spreading a baseband signal.
- Implement a DS-SS detection system to recover the original signal.
- Verify the functionality of the system through practical experiments and analysis.

1.2. Scope:

The scope of this project encompasses thorough research, design, simulation, and practical implementation of a DSSS communication system. The experimentation phase will focus on assessing the system's performance under varying conditions, providing valuable insights into the application of spread-spectrum techniques in real-world scenarios.

1.3. Methodology:

The methodology of this project follows a systematic approach encompassing the design, simulation, implementation, and experimentation phases. Each stage plays a crucial role in achieving the project's objectives.

- **Design:** Circuit design of a DSSS system, including a 4 MHz oscillator, carrier generation, pseudo-random sequence generation, data modulation, and BPSK-DSSS signal creation.
- **Simulation:** Software-based simulation to analyze the system under various conditions, guiding design refinements
- **Implementation:** Hardware construction based on schematics, considering real-world factors.
- **Experimentation:** Rigorous testing with an external PN sequence for synchronization, evaluating data recovery and real-world system performance.

1.4. Significance:

The significance of this project lies in bridging theoretical knowledge with practical implementation, providing a hands-on experience in working with spread-spectrum communication. The project's deliverables, including detailed circuit schematics, simulation results, and a fully functional DSSS system, will contribute to the educational resources available for students interested in this field. I anticipate that its significance will prompt consideration for inclusion as an official experiment in EE521 Communication Lab, given its potential to be mass-produced as a trainer board.

2. BACKGROUND OVERVIEW

2.1. The Concept of Spread Spectrum:

The conventional rule of assessing modulation scheme efficiency involves examining how efficiently it concentrates signal energy for a given information rate. While conventional wisdom favors signal compactness, spread-spectrum modulation techniques adopt the opposite approach—spreading the signal over a broad bandwidth.

Communication signals can be significantly expanded in bandwidth by factors ranging from 10 to 10,000 through the utilization of binary sequences and various techniques, as detailed later. This spreading yields **several advantages**:

- The signals gains immunity from various kinds of noise and multipath distortion. The earliest applications of spread spectrum were military, where it was used for its immunity to jamming.
- It can also be used for hiding and encrypting signals. Only a recipient who knows the spreading code can recover the encoded information.
- Several users can independently use the same higher bandwidth with very little interference. This property is used in cellular telephony applications, with a technique known as code division multiplexing (CDM) or code division multiple access (CDMA).

Figure 1. highlights the key characteristics of any spread spectrum system. Input is fed into a channel encoder that produces an analog signal with a relatively narrow bandwidth around some center frequency. This signal is further modulated using a sequence of digits known as a spreading code or spreading sequence. Typically, but not always, the spreading code is generated by a pseudonoise, or pseudorandom number, generator. The effect of this modulation is to significantly increase the bandwidth (spread the spectrum) of the signal to be transmitted. it dilutes the signal energy, resulting in minimal power density at any point within the spread signal, despite occupying a vast bandwidth. Signal dilution depends on factors such as transmitting power, distance from the transmitter, and spread signal width, potentially rendering the signal invisible to a conventional receiver but receivable with a spread-spectrum receiver.

On the receiving end, the same digit sequence is used to demodulate the spread spectrum signal. the receiver's ability to reject strong undesired signals, even those significantly stronger than the desired spread-spectrum signal power density. This rejection is facilitated by the receiver having a copy of the spreading sequence, which it uses to "despread" the signal, suppressing non-spread signals in the process.

Finally, the signal is fed into a channel decoder to recover the data.

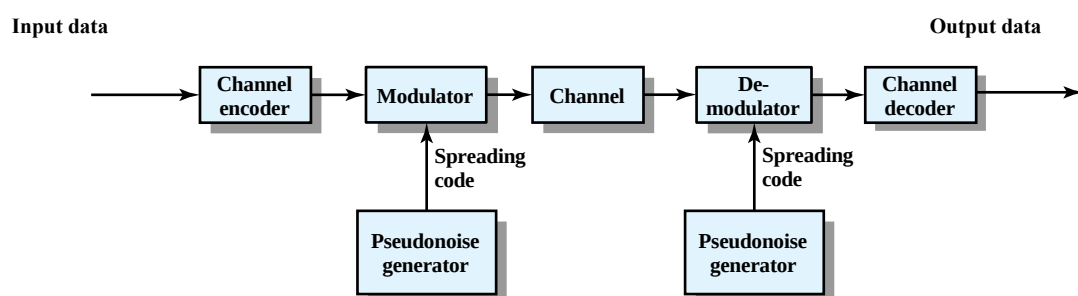


Figure 1. General Model of Spread Spectrum Digital Communication System

2.1.2. Types of Spread Spectrum

There are numerous ways to cause a carrier to spread; however, all spread-spectrum systems can be viewed as two modulation processes. First, the information to be transmitted is applied. A conventional form of modulation, either analog or digital, is commonly used for this step. Second, the carrier is modulated by the spreading code, causing it to spread out over a large bandwidth. Four spreading techniques are commonly used in communication systems:

1. Frequency Hopping (FH)

Frequency hopping involves rapidly altering the center frequency of a conventional carrier based on a pseudo-random list of channels. This process occurs at a high rate, with dwell times typically less than 10 milliseconds. The short dwell time minimizes interference both to and from conventional users.

2. Direct Sequence (DS)

Direct sequence spreading involves using a fast binary bit stream to shift the phase of an RF carrier. This binary sequence, known as pseudonoise (PN) or PN code, appears random but is generated digitally. DS spread spectrum is commonly used for transmitting digital information, and it employs techniques like binary phase-shift keying (BPSK) or quadrature phase-shift keying (QPSK).

3. Chirp Spread Spectrum

Chirp spread spectrum employs a carrier that sweeps over a range of frequencies. This technique is primarily used in ranging and radar systems.

4. Time Hopping

Time hopping involves on-off keying of the carrier by the PN sequence, resulting in a low-duty cycle signal. The speed of keying determines the amount of signal spreading.

*Hybrid Spread Spectrum Systems

Hybrid systems combine two or more spread spectrum techniques into a single system, often combining the advantages of each. The most common hybrid systems combine frequency hopping and direct sequence techniques, offering improved performance compared to single spread-spectrum techniques.

2.1.3. Spreading Sequences

A crucial element of spread spectrum technology is the Pseudonoise (PN) sequence responsible for spreading the RF carrier. The specific spreading sequence employed significantly influences the performance of spread spectrum systems. In frequency hopping, the spreading code comprises a sequence of numbers representing the channels for frequency hopping. Conversely, in a Direct Sequence (DS) system, the spreading code is a rapid bit stream generated by a digital circuit, designed to emulate random activity. Sequences with random-like properties are termed pseudorandom or pseudonoise sequences.

2.1.4. Correlation

Correlation, a fundamental process in spread-spectrum systems, serves as a common method for signal reception. It gauges the similarity between two signals, assigning a numerical value between zero and one, with one indicating a perfect match. In spread-spectrum receivers, correlation is frequently employed to identify signals encoded with a specific PN sequence.

Correlator Operation: A correlator, the circuit used for correlation, typically consists of a mixer followed by a low-pass filter for averaging. The mixer multiplies the two signals to be compared, generating an output value that indicates the degree of similarity. A high output signifies a strong match, while a lower output indicates dissimilarity. The averaging circuit reports the average likeness of the two signals over time.

Correlation in DS Systems: In DS systems, the correlator identifies and detects signals with the desired spreading code. Signals with different PN codes or those not spread at all yield lower outputs. The desired signal, having a strong match with the locally generated PN code, produces a larger correlator output. This process effectively rejects interference by distinguishing between desired and undesired signals.

Correlation in FH Systems: In frequency hopping systems, correlation operates differently but follows the same concept. The receiver, using the spreading sequence, synchronizes with the transmitter's carrier frequency, channel by channel. If the receiver loses synchronization, it cannot recover transmitted information. Signals under the influence of a different PN sequence are received randomly, minimizing interference, as they coincide with the desired signal only occasionally.

2.2. DIRECT SEQUENCE SPREAD SPECTRUM (DS-SS)

Direct Sequence Spread Spectrum (DSSS) systems, specifically those involving direct carrier modulation through code sequence modulation, stand out as the most recognized and extensively employed spread spectrum solutions. The prevalent adoption of DSSS can be attributed to its inherent simplicity, distinguishing it from other techniques. One key advantage lies in its operational efficiency, eliminating the need for intricate, high-speed, and fast-settling frequency synthesizers.

This simplicity not only streamlines the design and implementation process but also enhances the overall reliability and accessibility of DSSS systems. As a result, DSSS has become a go-to choice in various applications, showcasing its prominence in the realm of spread spectrum technologies.

With direct sequence spread spectrum (DSSS), each bit in the original signal is represented by multiple bits in the transmitted signal, using a spreading code. The spreading code spreads the signal across a wider frequency band in direct proportion to the number of bits used. Therefore, a 10-bit spreading code spreads the signal across a frequency band that is 10 times greater than a 1-bit spreading code.

One technique with direct sequence spread spectrum is to combine the digital information stream with the spreading code bit stream using an exclusive-OR (XOR). The XOR obeys the following rules:

$$0 \oplus 0 = 0 \quad 0 \oplus 1 = 1 \quad 1 \oplus 0 = 1 \quad 1 \oplus 1 = 0$$

Figure 2. shows an example. Note that an information bit of one inverts the spreading code bits in the combination, while an information bit of zero causes the spreading code bits to be transmitted without inversion. The combination bit stream has the data rate of the original spreading code sequence, so it has a wider bandwidth than the information stream. In this example, the spreading code bit stream is clocked at four times the information rate.

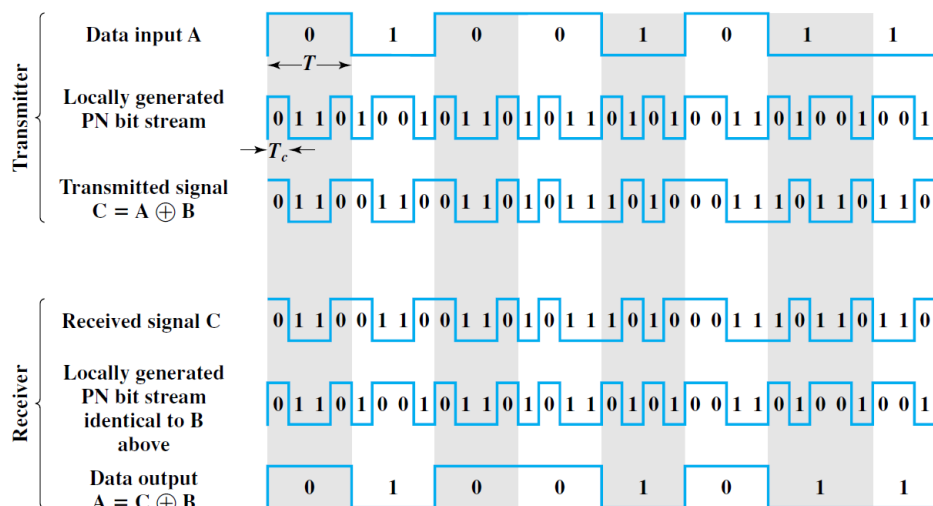


Figure 2 Example of Direct Sequence Spread Spectrum

2.2.1 DS-SS Using BPSK

To see how this technique works out in practice, assume that a BPSK modulation scheme is to be used. Rather than represent binary data with 1 and 0, it is more convenient for our purposes to use -1 and +1 to represent the two binary digits. In that case, a BPSK signal can be represented as was shown in Equation

$$s_d(t) = A d(t) \cos(2\pi f_c t)$$

where

A = amplitude of signal

f_c = carrier frequency

$d(t)$ = the discrete function that takes on the value of +1 for one bit time if the corresponding bit in the bit stream is 1 and the value of -1 for one bit time if the corresponding bit in the bit stream is 0

To produce the DSSS signal, we multiply the preceding by $c(t)$, which is the PN sequence taking on values of +1 and -1:

$$s(t) = A d(t)c(t) \cos(2\pi f_c t)$$

At the receiver, the incoming signal is multiplied again by $c(t)$. But $c(t) \times c(t) = 1$ and therefore the original signal is recovered:

$$s(t)c(t) = A d(t)c(t)c(t) \cos(2\pi f_c t) = s_d(t)$$

Equation (2) can be interpreted in two ways, leading to two different implementations. The first interpretation is to first multiply $d(t)$ and $c(t)$ together and then perform the BPSK modulation. That is the interpretation we have been discussing. Alternatively, we can first perform the BPSK modulation on the data stream $d(t)$ to generate the data signal $S_d(t)$. This signal can then be multiplied by $c(t)$.

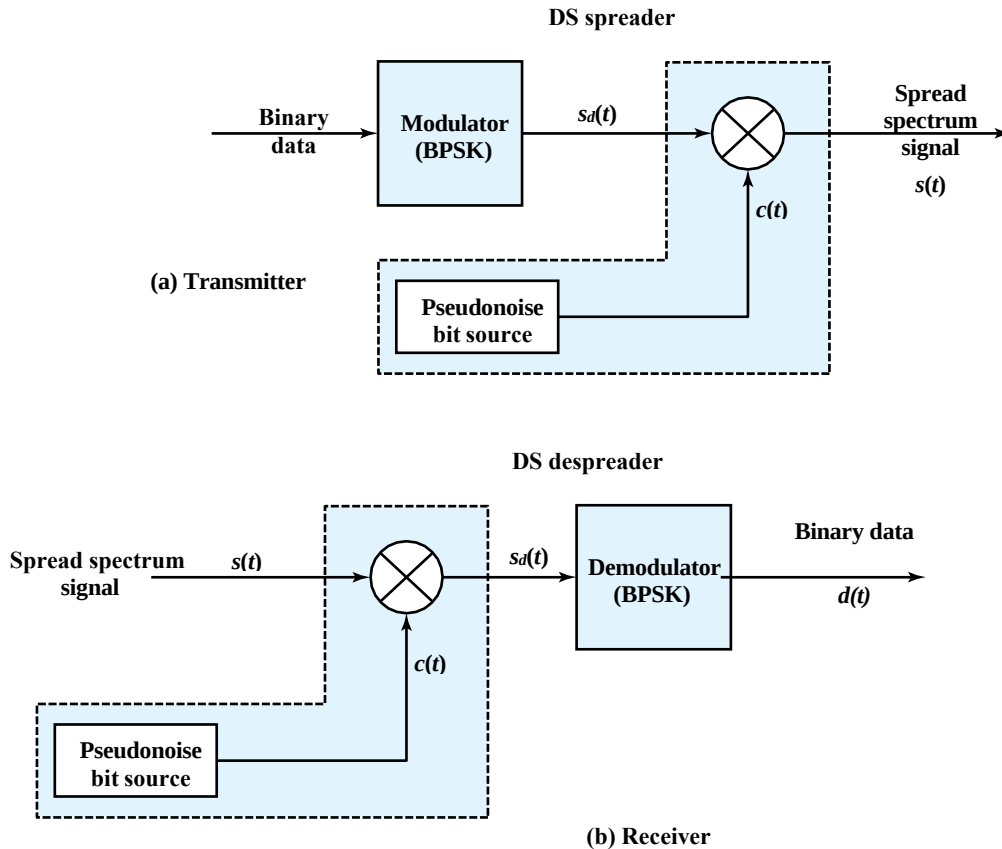


Figure 3. Direct Sequence Spread Spectrum System

An implementation using the second interpretation is shown in Figure 3. Figure 4. is an example of this approach.

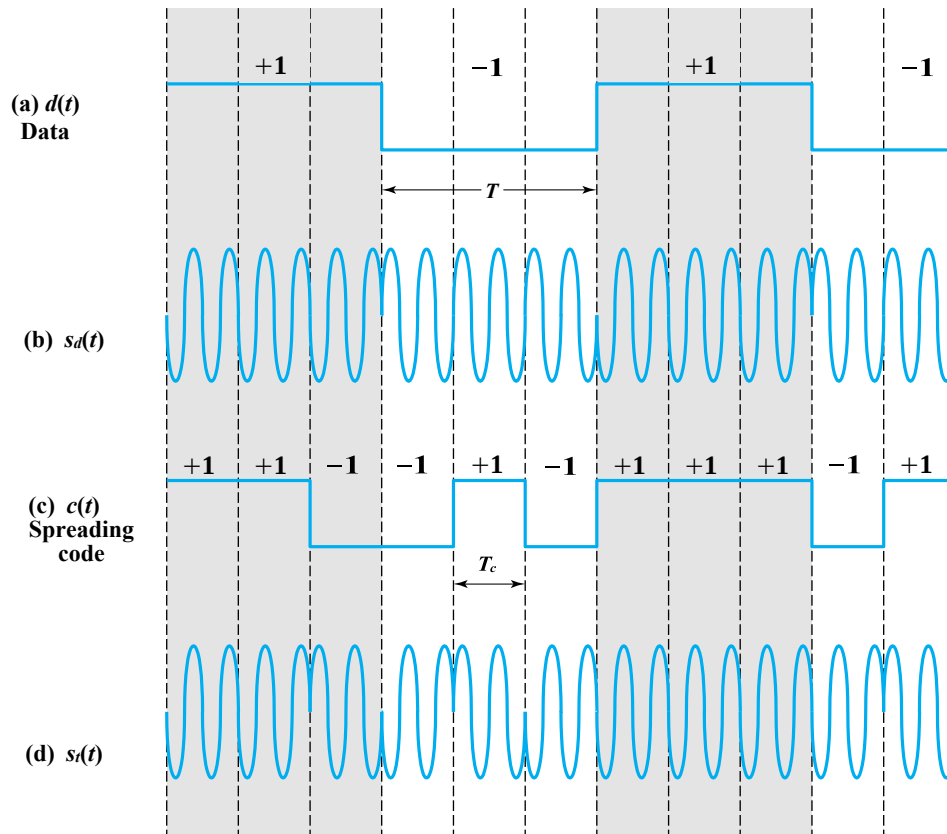


Figure 4 Example of Direct-Sequence Spread Spectrum Using BPSK

2.2.2. Spreading Sequence Generators

Generation of the PN sequence is commonly done with linear-feedback shift registers (LFSR). See Fig. 5. The shift register consists of a number of one-bit memory registers that shift their contents to the right with each clock pulse. New values are introduced at the left-most stage while output is commonly taken from the right-most stage.

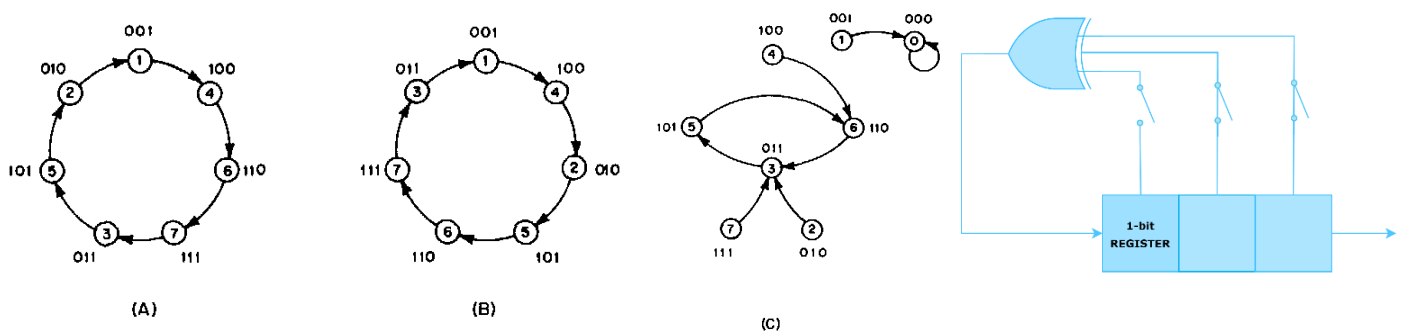


Figure 5 Various PN sequences are depicted in the diagram. At position A, a three-stage shift register with (1,3) stages tapped generates seven unique code words, each the width of the shift register. Position B shows a three-stage shift register with (2,3) stages tapped, resulting in a different arrangement of code words. At position C, the choice of taps (1,2) for the three-stage register produces two non-maximal-length cycles. Both positions A and B exhibit maximal-length cycles. Position D features a three-stage linear feedback shift register where the stages for feedback are selected by S1-S3 to be combined using exclusive-OR.

The shift register can be used to generate a wide variety of sequences. Generation of a linear sequence is typically done by tapping the values of certain cells and combining the values using an exclusive-or (one-bit addition without carry) gate. The combined output is then fed back to the input of the left-most cell.

The LFSR sequence is a repeating sequence whose maximum length is related to the number of stages (N) in the shift register by the equation: $\text{length} = 2^N - 1$. This equation determines the maximum length; not all tap choices produce the longest sequence, however. There are two cases to consider: the maximal-length sequence or m-sequence and the nonmaximal-length or composite sequences.

The length of an m-sequence is given by the equation $(2^N - 1)$ and is the longest possible for a given register length. Since the output repeats after producing $(2^N - 1)$ bits, the entire sequence can be viewed as a cycle (see Fig. 5). Initializing the register with any sequence of consecutive bits from the cycle will generate the entire cycle from that point. A cycle is commonly started by an initial value of all zeros followed by a one such as 000001 for a seven-stage register.

Other starting values still generate the same sequence but are said to be phase shifted from the zeros followed by a one initial starting point. A run of bits selected from the sequence that is the same length as the generating shift register is called a code word of the sequence. All code words from an m-sequence are unique with no repeats, and no code word contains all zeros.

Comparatively, the length of the nonmaximal sequence is always shorter than $(2^N - 1)$, and its outputs form a number of distinct short cycles. Code words are not unique for this type of sequence. The nonmaximal sequences are called a composite sequence because a composite sequence can be expressed as a combination of shorter maximal-length sequences. The nonmaximal-sequence length is in fact the product of the length of their smaller m-sequences of which it is composed. Determining whether a given tap sequence will produce a maximal or composite sequence is quite complicated. Most efforts have been carried out by computer search methods, and a number of excellent tables are available for the designer. A maximal-length sequence seven-stage shift register can easily be built with two ICs, as shown in Fig. 20. A more complete description of this shift register.

▪ **Orthogonality**

An important consideration in choosing a sequence for a spread-spectrum system is the amount of statistical similarity a sequence has with conventional signals and with sequences employed by other spread-spectrum systems. The greater the degree of similarity, the less able the spread-spectrum receiver will be able to reject interference. The ideal sequence will show a very low correlation when compared with undesired sequences. Moreover, the PN sequence should also have a low correlation with shifted versions of itself. A measure of a sequence's self-correlation is measured by its autocorrelation function (ACF). The ACF measures how much interference will be received from other spread-spectrum units using the same PN sequence but with different starting points within the PN code cycle. Two code families that have found common usage are the m-sequences and Gold codes. Gold codes are a family of spreading codes generated by exclusive-OR combining the output of two "preferred" m-sequence generators. Many other PN codes can be used with spread spectrum. Dixon(1.....) lists a number of such codes.

In general the used codes should have the following **properties**:

○ **Balance property**

The occurrence of 0 and 1 in the sequence should be approximately the same. More precisely, in a maximum length sequence of length $(2^N - 1)$ there are (2^{N-1}) ones and $(2^{N-1} - 1)$ zeros. The number of ones equals the number of zeros plus one, since the state containing only zeros cannot occur.

- **Run property**

A "run" is a sub-sequence of consecutive "1"s or consecutive "0"s within the MLS concerned. The number of runs is the number of such sub-sequences.

Of all the "runs" (consisting of "1"s or "0"s) in the sequence :

- One half of the runs are of length 1.
- One quarter of the runs are of length 2.
- One eighth of the runs are of length 3. ... etc. ...

- **Correlation property**

The circular [autocorrelation](#) of an MLS is a [Kronecker delta](#) function (with DC offset and time delay, depending on implementation). For the ± 1 convention, i.e., bit value 1 is assigned $s = +1$ and bit value 0 $s = -1$, mapping XOR to the negative of the product:

$$R(n) = \frac{1}{N} \sum_{m=1}^N s[m] s^*[m+n]_N = \begin{cases} 1 & \text{if } n = 0, \\ -\frac{1}{N} & \text{if } 0 < n < N. \end{cases}$$

where s^* represents the complex conjugate and $[m+n]_N$ represents a [circular shift](#). The linear autocorrelation of an MLS approximates a Kronecker delta.

- **Synchronization**

Synchronization is the most difficult issue for spread-spectrum systems. For a spread-spectrum receiver to demodulate the desired signal, it must be able to synchronize the locally generated PN code reference with the one used by the transmitter. This operation commonly takes place at very high speeds. It is usually viewed as two processes: rough synchronization which searches and acquires the receiver's PN code within one bit or channel of the transmitter's, and fine synchronization which maintains bit or channel dwell timing. In rough synchronization, the receiver attempts to line up the local PN code as close as possible to the transmitter's. There are two methods. The first is called epoch synchronization, in which the transmitter periodically sends a special synchronization sequence. Commonly, this synchronization sequence is a unique short bit sequence chosen because it is easily detected by a simple correlator known as a digital matched filter. This filter consists of a shift register that clocks the received sequence by one bit at a time and compares it against the unique sequence. The output of the matched filter is highest when the unique word is found in the input bit stream. A second method of synchronization is known as phase synchronization. Here, the receiver attempts to determine which of the $(2^N - 1)$ phases the m-sequence could be in. To determine the phase, and thereby the synchronization, at least N bits would have to be received without error. Noisy signals may require more than N bits depending on the signal-to-noise ratio. PN codes with short cycle times will exhibit several repetitions in a short amount of time. A digital matched filter can be constructed to signal the presence of a unique code word from this short spreading code sequence. This is essentially a form of epoch synchronization. Longer sequences require more-complex synchronization methods. One method uses a sequence based on the time of day. The receiver sets up a digital matched filter with a value several seconds ahead of the current time and waits for this unique code word to appear. The procedure may be useful for net operations in which stations can enter or leave a net at will. There are a number of specially developed sequences available for rapid synchronization; however, they fall beyond the scope of this brief discussion.

2.2.3. DSSS Performance Considerations

The spectrum spreading achieved by the direct sequence technique is easily determined (Figure 9.9). In our example, the information signal has a bit width of T , which is equivalent to a data rate of $1/T$. In that case, the spectrum of the signal, depending on the encoding technique, is roughly $2/T$. Similarly, the spectrum of the PN signal is $2/T_c$. **Figure6**

shows the resulting spectrum spreading. The amount of spreading that is achieved is a direct result of the data rate of the PN stream.

As with FHSS, we can get some insight into the performance of DSSS by looking at its effectiveness against jamming. Let us assume a simple jamming signal at the center frequency of the DSSS system. The jamming signal has the form:

$$s_j(t) = \sqrt{2S_j} \cos(2\pi f_c t)$$

and the received signal is

$$s_r(t) = s(t) + s_j(t) + n(t)$$

where

$s(t)$ = transmitted signal

$s_j(t)$ = jamming signal

$n(t)$ = additive white noise

S_j = jammer signal power

The despreader at the receiver multiplies $s_r(t)$ by $c(t)$, so the signal component due to the jamming signal is

This is simply a BPSK modulation of the carrier tone. Thus, the carrier power S_j is spread over a bandwidth of approximately $2/T_c$. However, the BPSK demodulator (Figure 3) following the DS-SS despreader includes a bandpass filter matched to the BPSK data, with bandwidth of $2/T$. Thus, most of the jamming power is filtered.

Although a number of factors come into play, as an approximation, we can say that the jamming power passed by the filter is

$$S_{jF} = S_j(2/T)/(2/T_c) = S_j(T_c/T)$$

The jamming power has been reduced by a factor of T_c/T through the use of spread spectrum. The inverse of this factor is the gain in signal-to-noise ratio:

$$G_P = \frac{T}{T_c} = \frac{R_c}{R} \approx \frac{W_s}{W_d}$$

where R_c is the spreading bit rate, R is the data rate, W_d is the signal bandwidth, and W_s is the spread spectrum signal bandwidth.

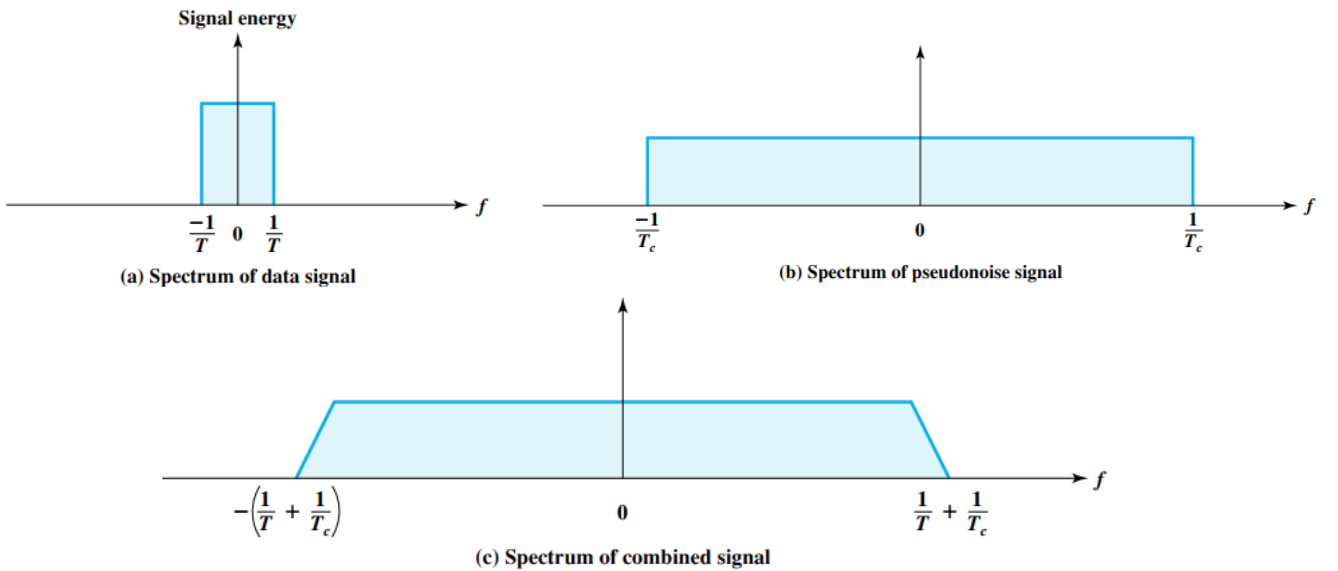


Figure 6 Approximate Spectrum of Direct Sequence Spread Spectrum Signal

3. SYSTEM DESIGN AND DEVELOPMENT: A DSSS COMMUNICATION LINK

Referring to Figure 6, conceptualizing an amateur spread-spectrum communication link becomes relatively straightforward when leveraging an existing external reference, such a crystal oscillator or radio transmitter, is used. In this experiment, the external reference is a 4-MHz signal obtained from a function generator. A 2-MHz carrier is obtained by dividing the reference frequency by 2; chosen for its easy observation of DS-SS phenomena on standard low-frequency oscilloscopes and proximity to practical HF systems. this carrier is also used as clock for a pseudo-random generator. Data is XORed with the output of the pseudo-random generator and the result is XORed with the carrier. This produces a binary phase shift keyed direct sequence spread spectrum (BPSK DSSS) signal. At the receiving site, a pseudo-noise sequence similar to that created at the transmitter, is generated and synchronized to the original one. By mixing this sequence with the reference carrier, it is possible to extract the original data. In my demonstration model, points T and R were connected to each other with a jumper. In a practical application, point T would be connected to an antenna via an amplifier; point R would similarly be connected to a receiving antenna, AGC amplifier and the like.

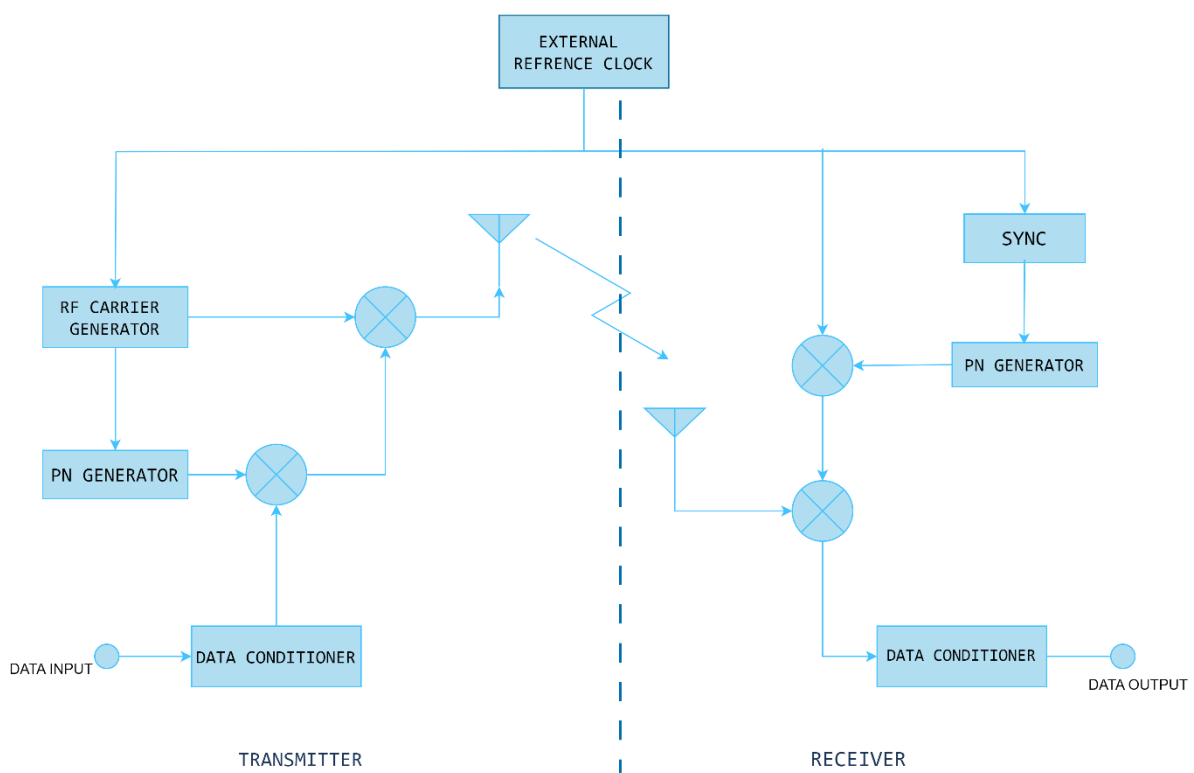


Figure 7 A block diagram of a possible radio simple spread-spectrum communications link.

3.1. Circuit Design & Operation

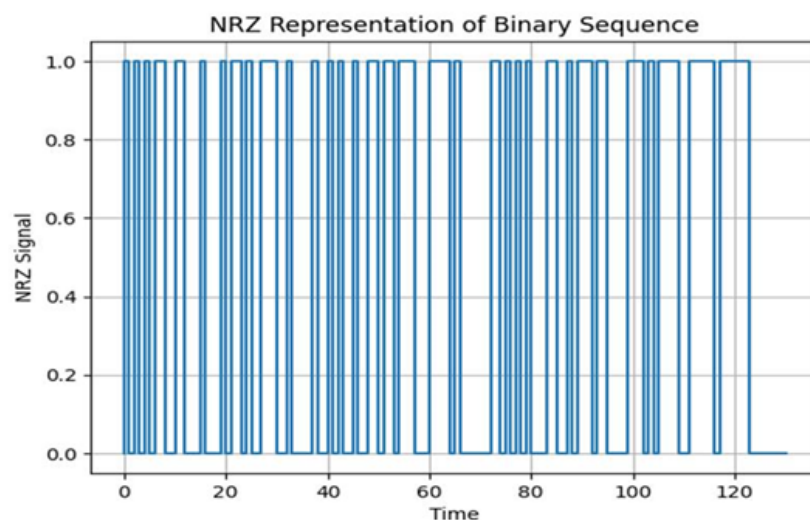
3.1.1. PN Generator:

The PN code used in the Transmitter and receiver stage A **maximum length sequence (MLS)** is a type of [pseudorandom binary sequence](#). Using 7-stage shift register The 74164 is an 7-stage shift register, and the 7486 is a quad XOR gate. One section of the 7486 is used to combine the feedback from taps 1 and 7, while another section is used as an inverter. The purpose of the inverter can be

The pseudo-random sequence generated by this described arrangement is shown in Table 1. The numbers listed are the decimal equivalents of the binary states appearing at the 74164's output pins

Table 1
Pseudo-random Sequence Generated by the
Described Circuit

Table 1						0001110	14	0110001	49	1111010	122
Pseudo-random Sequence Generated by the Described Circuit						1000111	71	0011000	24	0111101	61
						1100011	99	1001100	76	0011110	30
						1110001	113	0100110	38	1001111	79
0000000	0	1000100	68	0010000	16	1111000	120	1010011	83	1100111	103
1000000	64	0100010	34	1001000	72	0111100	60	1101001	105	1110011	115
0100000	32	1010001	81	0100100	36	1011110	94	1110100	116	1111001	121
1010000	80	1101000	104	1010010	82	0101111	47	0111010	58	1111100	124
0101000	40	0110100	52	0101001	41	0010111	23	1011101	93	0111110	62
1010100	84	1011010	90	0010100	20	0001011	11	1101110	110	1011111	95
0101010	42	0101101	45	1001010	74	0000101	05	0110111	55	1101111	111
1010101	85	0010110	22	0100101	37	0000010	02	0011011	27	1110111	119
1101010	106	1001011	75	0010010	18	1000001	65	0001101	13	1111011	123
0110101	53	1100101	101	1001001	73	1100000	96	0000110	06	1111101	125
0011010	26	1110010	114	1100100	100	0110000	48	1000011	67	1111110	126
1001101	77	0111001	57	0110010	50	1011000	88	1100001	97	0111111	63
1100110	102	0011100	28	1011001	89	0101100	44	1110000	112	0011111	31
0110011	51	1001110	78	1101100	108	1010110	86	0111000	56	0001111	15
0011001	25	0100111	39	0110110	54	0101011	43	1011100	92	0000111	07
0001100	12	0010011	19	1011011	91	0010101	21	0101110	46	0000011	03
1000110	70	0001001	09	1101101	109	0001010	10	1010111	87	0000001	01
0100011	35	0000100	04	1110110	118	1000101	69	1101011	107	0000000	00
0010001	17	1000010	66	0111011	59	1100010	98	1110101	117	→Repeat Sequence	
0001000	08	0100001	33	0011101	29						



3.1.2. Transmitter Stage Description and Operation

Reviewing the transmitter stage in Fig 8, implements a BPSK Direct Sequence Spread Spectrum (DSSS) transmitter.

Reference Frequency and Carrier Generation

U1 is a 4-MHz DIP oscillator serving as the reference frequency. U2A, a 4013 D flip-flop, divides this frequency by 2 to generate a 2-MHz carrier (referred to as **CA**). Subsequently, the 2-MHz frequency is further divided by 1,000 through three consecutive 4017 divide-by-10 stages, U4, U5, and U6, to produce a 2-kHz square wave. This 2-kHz signal serves as both a test square wave and as the clock for another 4013 D flip-flop, U2B.

Data input

Under normal conditions, when the Morse key is open, the D input of U2B remains low, causing its Q output to be low as well. Upon pressing the key, the D input goes high, resulting in Q transitioning to +5 V at the next clock pulse. Releasing the key causes the D input to return low, and subsequently, Q also transitions low at the next clock pulse. Consequently, the data (in this case, the Morse sequence) becomes synchronized to the 2-kHz clock, which is a submultiple of the 2-MHz carrier. This synchronization ensures phase-coherent switching. The Q output of U2B serves as our data, denoted as **DA**.

Spreading

Additionally, the 2-MHz carrier (**CA**) is utilized to clock U7, a 74164 7-stage linear feedback shift register configured as a pseudo-random noise generator (PN), with feedback taps at outputs 1 and 7. Feedback is applied using an XOR stage (U8B), while U8A functions as an inverter. The PN sequence, available at pin 2 of U7 and referred to as the original code (**OCO**), is modulo-2 added to the data output of U2B in an XOR stage, U8C. The resulting signal, **OCO x DA**, is then XORed with the 2-MHz carrier (**CA** in U8D) to produce the output signal, **OCO x DA x CA**, representing a Binary Phase Shift Keyed Direct Sequence Spread Spectrum (BPSK DSSS) signal.

In practical applications, this signal would typically be applied to an antenna using a power amplifier. However, on the experimental board, it is directly jumpered to the input of the receiver.

3.1.3. Transmitter Stage Description and Operation

Reviewing the receiver stage, we observe that the input signals **OCO x DA x CA** and **CA** (the carrier derived from the external reference) are both fed into the inputs of an XOR stage, U17D. Initially, at the start of transmission when no data is transmitted **DA=0**, the XOR operation between **OCO x DA x CA = OCO x CA** and **CA** results in **OCO**, representing the original code sequence. This sequence is then loaded serially into U13, a 7-stage shift register clocked by **CA**. This shift register acts as a serial to parallel converter that takes the serial data stream as input and then outputs the data in parallel format, with each bit appearing on a separate output line.

Additionally, another shift-register stage, U20, functions as a PN generator, employing the same pseudo-noise sequence as the transmitter, its output in which denoted (**LCO**). To ensure synchronization between both operators of the radio link, it is crucial to clearly announce the type of register used in spread spectrum. When the receiver is activated, U20 operates at twice the normal 2-MHz rate, achieved by applying the 2-MHz carrier (**2 x CA**) to U15D, a multiply-by-two stage. **CA** is Connected directly to one input of the XOR gate and connect it to the other input via a delay network. The delay network can be just a resistor in series with the signal and a capacitor to ground from the gate input end of the resistor. The output is then shaped by a Schmitt trigger (U19A, U19B).

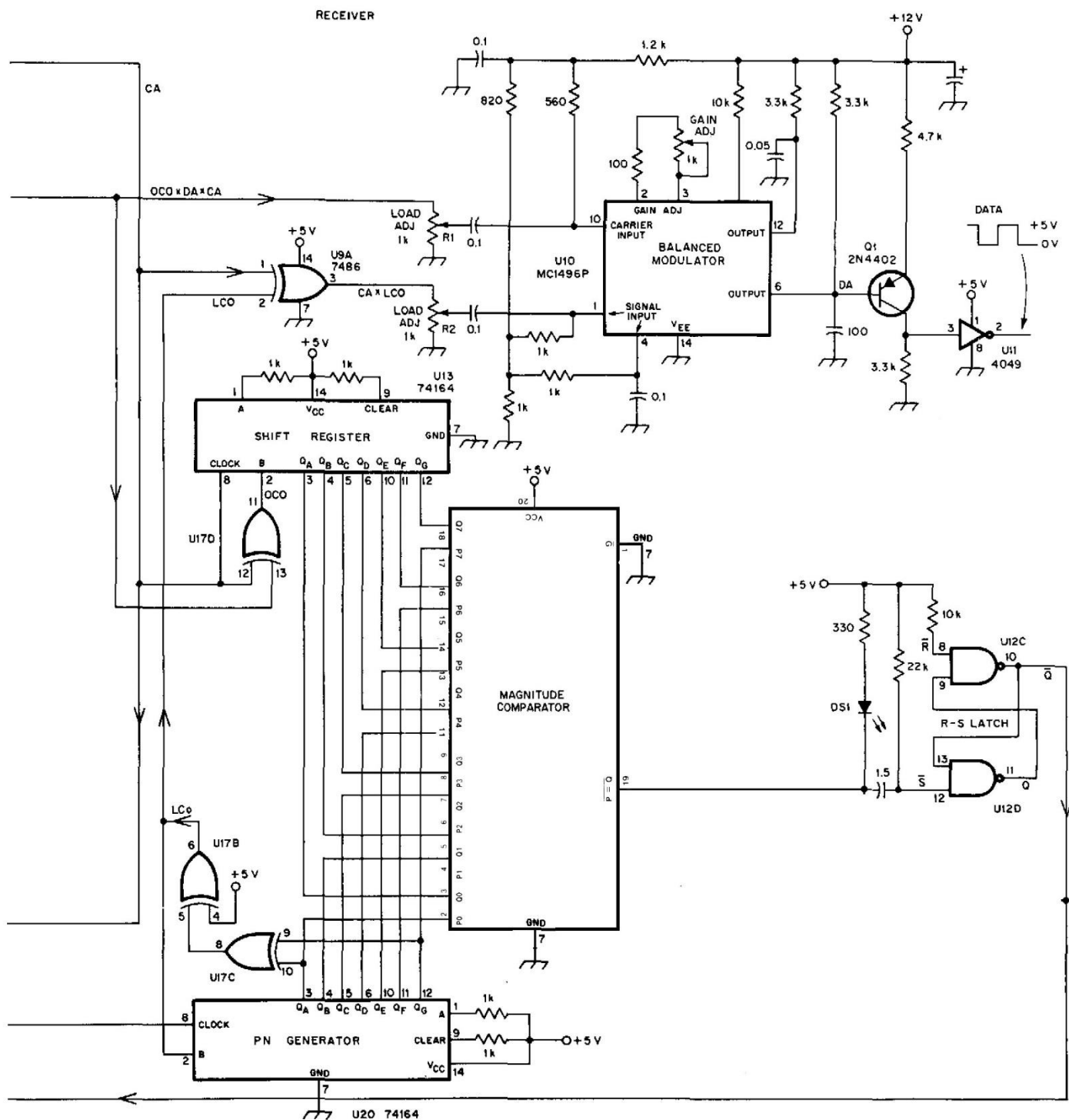


Figure 8 —A complete schematic diagram of the DSSS transmitter and receiver stages. Operation of the circuit is explained in the text.

U18A and U18B (part of a 4066-quad switch) are connected as a SPDT switch. U18B is normally closed, U18A is normally open and the 4-MHz clock signal is applied to the PN generator U20.

Since both U8 and U20 utilize the same feedback taps, they produce identical random sequences. The objective is to synchronize both signals to be in phase. Initially, when no data is being transmitted, the output of U13 represents **OCO**, the original code sequence. The outputs of U13 and U20 are compared bit by bit in a magnitude comparator 74688.

As U20 operates at twice the rate of U13, its output attempts to catch up with that of U13. There exists a moment when both output words are identical bit by bit. At this instant, The output of the magnitude comparator goes low. This action illuminates the "lock" LED and lowers the Q output of U12C. Consequently, U18B opens and U18A closes, switching the clock rate of U20 from 4 MHz to 2 MHz. Subsequently, both U7 and U20 produce the same PN sequences, are clocked at the same frequency, and are in phase (**OCO=LCO**).

To prevent further changes at the output of the magnitude comparator from affecting the clock speed, something that would otherwise happen whenever data is **received**, an edge-triggered RS latch (U12C, D) is utilized.

During transmission of Morse code, the lock LED may extinguish, indicating that the code extracted from the DSSS transmission (**OCO x DA x CA** XORed with **CA** which is loaded serially into U13) is no longer in phase with either **OCO** or **LCO**, The latter two, however, are still in phase. , and as soon as the key is released, the LED lights again.

Demodulation and recovery

The carrier (**CA**) and **LCO** are then XORed in U9A, with the output (**CA x LCO**) applied to one of U10's inputs, a balanced demodulator (LM1496P/1596). The other input receives the received DSSS signal (**OCO x DA x CA**). Due to the equivalence of (**OCO=LCO**), the demodulator's output is the data DADA. To adjust the three potentiometers associated with U10, an oscilloscope should be connected to U10, pin 6. As U10 operates from +12 V, a DC voltage shifter composed of Q1 and U11A is employed to shift the data to TTL levels.

With the design of the link complete, it's ready to be tested virtually to ensure correct operation, which is the purpose of the next section.

4. DIGITAL SIMULATION FOR DSSS SYSTEM VERIFICATION

Prior to constructing the physical DSSS communication link, a comprehensive simulation was conducted to ensure design functionality and identify potential areas for refinement. This simulation phase facilitated pre-implementation validation, reducing risks, and optimizing the final circuit design.

4.1. Simulation Environment

- Simulations were performed using **NI Multisim. Multisim™ Electronics Workbench™** is industry-standard SPICE simulation and circuit design software for analog, digital, and power electronics in education and research. Multisim is widely used in academia and industry for circuits education, electronic schematic design, and SPICE simulation. It was introduced in EE219 Basic Electronics course. This software was selected due to these key advantages (was chosen for its extensive built-in library of TTL and CMOS gates and, importantly, the availability of a working SPICE model for the balanced demodulator component).:
 - **Extensive Component Library:** Access to a wide array of TTL and CMOS gates simplified the modeling process.
 - **Balanced Demodulator Model (MC/LM1496):** The availability of a working SPICE model for this critical component was essential for accurate simulation.

4.2. Circuit Model

- A detailed model of the DSSS circuit was created within the simulation environment, faithfully replicating the components and connections of the design schematic (see Design Section).

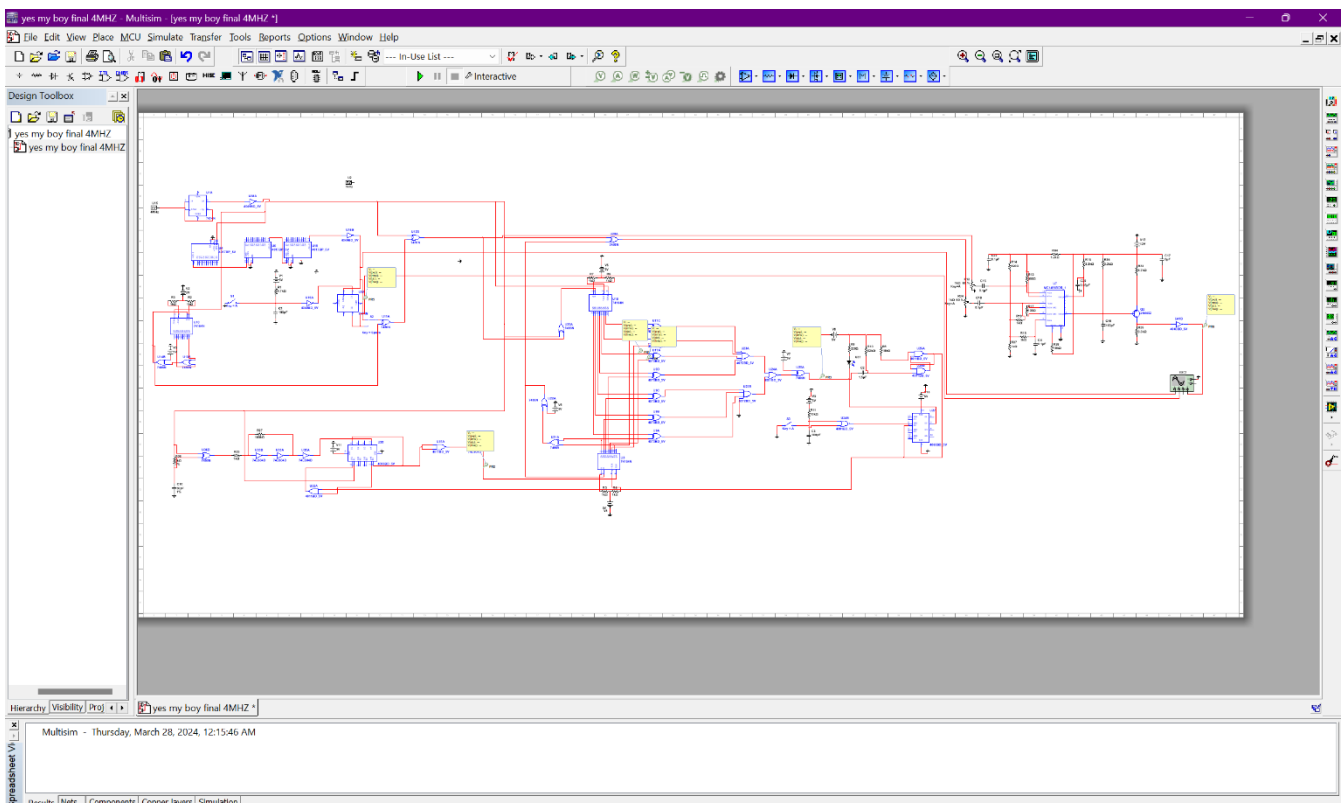


Figure 9. Screenshot from inside Multisim simulation

4.3. Simulation Results:

Figure 10. Oscilloscope waveform of different stages at Transmitter & Receiver



**At Transmitter
Data input (DA)**

**At Receiver
Recovered Data**

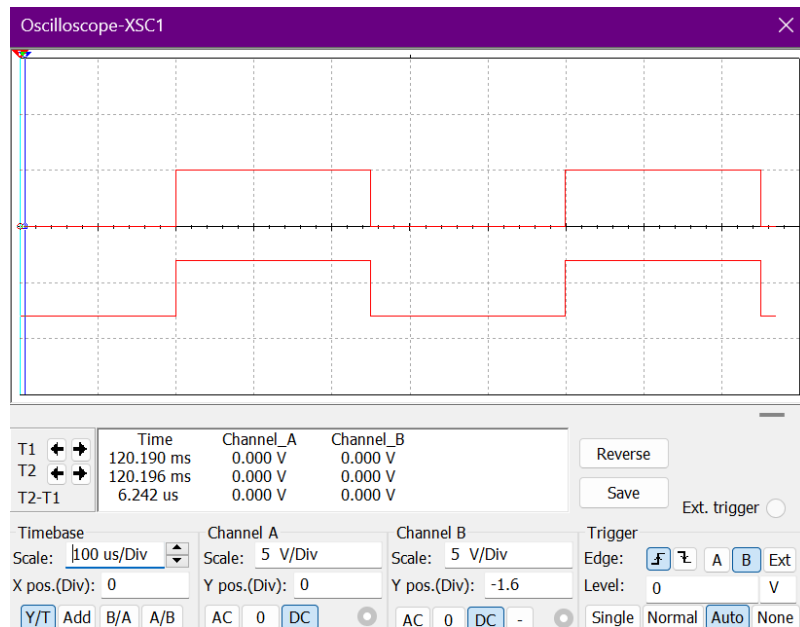
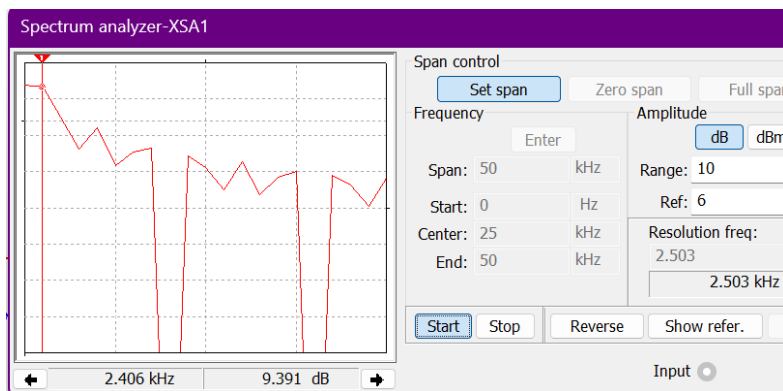
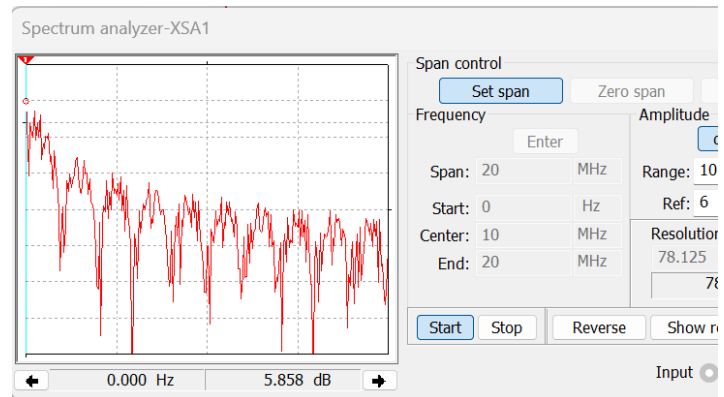


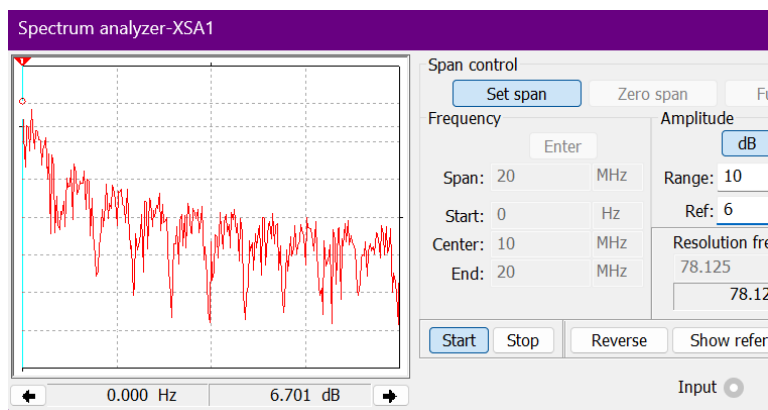
Figure 11. Spectrum Analyzer waveform of different stages at Transmitter & Receiver



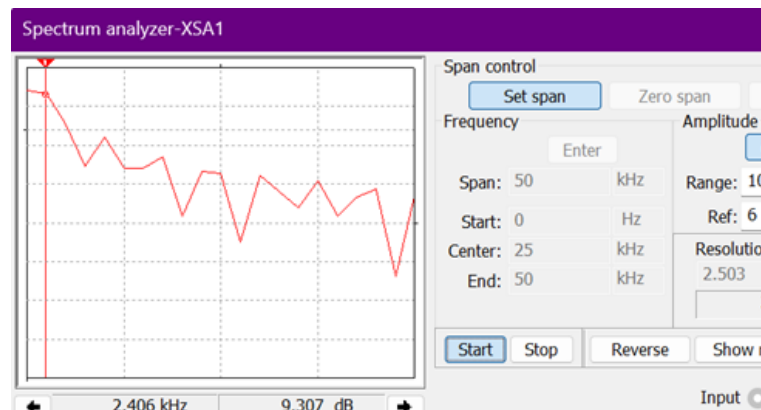
a)Spectrum of Data input 2KHz (DA)



b)Spectrum of Spreaded Data (OCO x



c)Spectrum of BPSK Signal(OCO x DA x CA)



d)Spectrum of deSpreaded (recovered)

The simulation phase successfully validated the theoretical expectations of the DSSS system, confirming its functionality before proceeding with its physical realization.

5. HARDWARE IMPLEMENTATION OF THE DSSS LINK:

Following the successful simulation phase, the DSSS communication link was initially implemented using a breadboard approach for rapid prototyping and testing. Future iterations will involve designing a PCB for a more robust and compact implementation.

5.1. Method

The DSSS communication link was realized using breadboards and readily available DIP (Dual In-Line Package) integrated circuits. This approach allowed for quick assembly and flexibility during the subsequent testing and verification phase. Further development will include designing a PCB for a more permanent and robust implementation. While PCB design software will be used in the next stage, the initial breadboard implementation did not involve any PCB-specific software.

5.2. Components and Sourcing

A Bill of Quantity and Cost (Budget) is included below, specifying all components used in the breadboard implementation of the DSSS system. Components were sourced from:

- **Power of Digital Electronics (PODE) store at Tarasan Street**
- **Electronic Service Center at Al Dahra Electronics Street**

Tabel 2. Bill of Quantity and Cost (Budget)

Component Type	Specific Part Number	Quantity	Cost per Unit (LYD)	Total Cost (LYD)
PNP transistor	2N4402	1	3.00	3.00
Dual D flip-flop	CD4013BC	1	3.00	3.00
Decade counter/divider	CD4017BC	2	5.00	15.00
Hex inverting buffer	CD4049BC	3	3.00	3.00
8-bit serial-in parallel-out shift register	74LS164	3	3.00	9.00
quad 2-Input exclusive OR gate	74LS86	3	3.00	9.00
hex Inverter	74LS04	4	3.00	12.00
quad 2-Input NAND . gate.	CD4011	1	3.00	3.00
quad bilateral switch	CD4066BC	1	3.00	3.00
Balanced Modulator-Demodulator	LM1496	1	16.00	16.00
8-Bit Magnitude Comparator	74HC688	1	7.00	7.00
Normally open push-button switch	-	1	3.00	3.00
Panel Mount potentiometer	1K Ω potentiometer, linear taper	2	1.50	3.00

LED	Green LED 2 Pins circular	1	2.00	2.00
Breadboard	Solderless breadboard MB-102	3	5.00	15.00
Jumper wires	Male-to-male jumper wires Set	40 set	5.00	5.00
Resistors	4.7k Ω x3 1k Ω x9 10k Ω x4 330 Ω x1 22k Ω x1 560 Ω x1 820 Ω x1 3.3k Ω x3 100 Ω x1 3k Ω x1 100k Ω x1 1.2k Ω x1 22 k Ω x1	34	0.200	7.00
Capacitors	100 pF ceramic capacitor x4 1.5 μ F ceramic capacitor x1 0.1 μ F ceramic capacitor x5 50 pF ceramic capacitor x1 6 μ F tantalum capacitor x1 0.05 μ F ceramic capacito x1	12	0.500	6.00
Total Cost (LYD)				115.00

5.3. Assembly Process

The assembly of the DSSS circuit on a breadboard was a straightforward process due to the clear schematics and simulation results, The following steps were taken:

1. **Component Preparation:** All components were checked against the Bill of Materials, and any leads were pre-shaped for breadboard insertion.
 2. **Power Rails:** +5V and ground rails were established along the length of the breadboard.
 3. **IC Placement:** Integrated circuits were carefully placed on the breadboard, ensuring correct orientation and pin spacing.
 4. **Passive Components:** Resistors and capacitors were added according to their values, paying attention to polarity on polarized capacitors.
 5. **Wiring:** Jumper wires were used to create the connections outlined in the schematic. Wiring was double-checked at each stage for accuracy.
- **Component Verification:** Before placing each component on the breadboard, its pinout was carefully verified against the relevant datasheet to ensure correct connections and avoid potential damage.

Due to the complexity of the DSSS system, the design was implemented on three separate breadboards to enhance modularity and facilitate troubleshooting. Each breadboard was functionally tested before integrating them into the complete system.

- **Board 1: Transmitter** - This breadboard housed all the components responsible for generating the DSSS signal, including [7486 XOR gates, 7404 NOT gates , D-Flip Flop 4013, Data input

pushbutton, LSFR 74164 IC]. After assembly, the functionality of the transmitter board was verified by measuring the generated signal using an oscilloscope.

- **Board 2: Receiver (PN Generator and Synchronization Circuit)** - The second breadboard focused on the receiving functionality. It included a pseudo-noise (PN) generator circuit to match the transmitter's sequence, along with components for synchronization. Once assembled, this board's operation was tested by feeding a pre-recorded DSSS signal from an external source and observing the output on the oscilloscope for synchronization verification.
- **Board 3: Balanced Demodulator and Synchronization (Partial)** - The third breadboard contained the balanced demodulator circuit responsible for recovering the original data from the received DSSS signal. It also included part of the synchronization circuit, specifically an RS latch with an LED indicator. Successful synchronization was indicated by the LED illumination upon receiving a compatible PN sequence.

This modular approach to assembly minimized troubleshooting efforts and allowed for individual verification of each functional block before integrating them into the complete DSSS system.

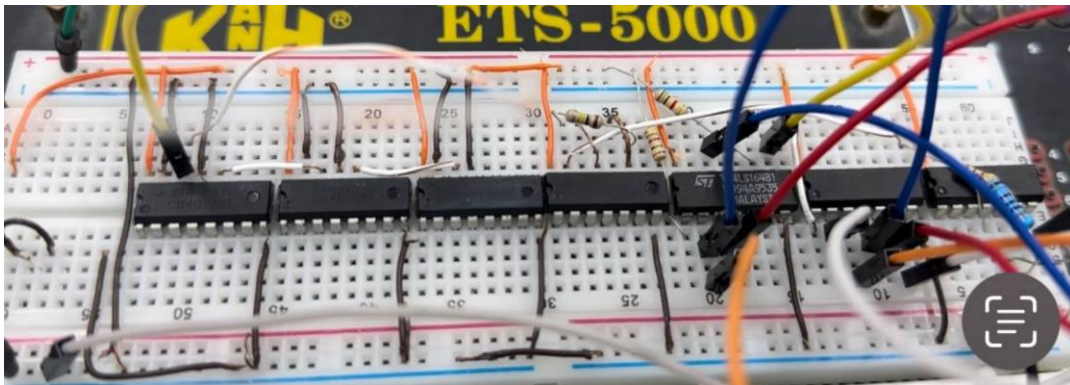


Figure 12. Breadboard 1 containing the DSSS transmitter

- **Assembly Stability:** To ensure stability during testing, the three breadboards were secured onto a platform using double-sided tape.

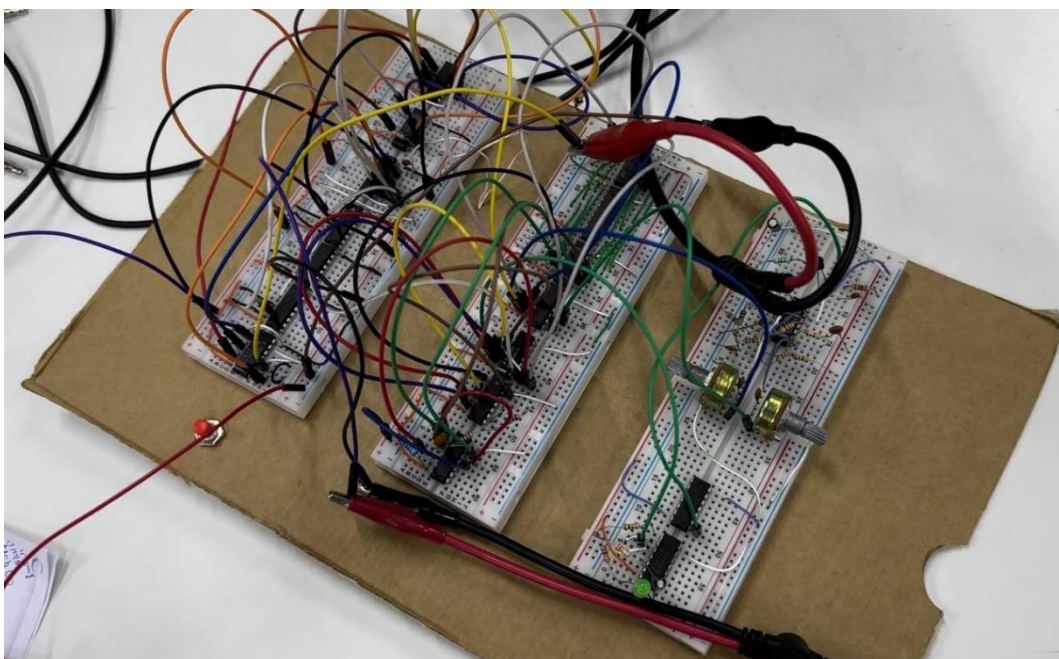


Figure 13. Breadboard assembly of the DSSS system, consisting of three interconnected breadboards to accommodate transmitter and receiver circuitry.

5.4. Challenges

- **Compactness:** A primary challenge was the sheer number of components required for the DSSS system. To maintain a tidy and compact layout, maximum use was made of each IC's available inputs. However, this led to complex and tightly spaced wiring.
- **Loose Connections:** One minor challenge encountered was an initially intermittent connection caused by a slightly loose jumper wire. This was resolved by carefully verifying all connections and ensuring a snug fit in the breadboard.
- **Overcoming Wiring Challenges:** To manage the tangled wires and complexity, single-conductor AWG 24 wires were obtained by stripping a CAT5 cable. These were used to neatly distribute VCC (5V) and GND connections to each IC, reducing clutter and improving visual clarity.
- **Component Performance:** Several components did not perform according to their datasheet specifications, especially when operating at the higher frequencies (MHz range) required for this project. This added an extra layer of complexity during troubleshooting efforts.
- **Faulty Components:** A few components were found to be faulty, creating additional challenges during the debugging process.
- **Component Sourcing:** Obtaining certain specialized components, such as the decade counter IC and the LM1496 balanced demodulator, proved difficult within Libya.

With the breadboard implementation of the DSSS system complete, the next step is to perform rigorous testing and validation of its functionality. This testing will focus on verifying PN sequence generation, synchronization accuracy, demodulation effectiveness, and overall system.

6. DSSS SYSTEM EVALUATION AND PERFORMANCE TESTING

With the DSSS system successfully constructed, a comprehensive testing program was undertaken to validate its functionality, verify performance, and identify potential areas for improvement. This testing phase followed a systematic approach, addressing the following key objectives:

1. **Verification of PN Sequence Generation:** Initial tests focused on confirming that the transmitter and receiver's PN generators produced the expected sequence as per the chosen configuration. An oscilloscope was used to visualize the output of both PN generators for:
 - **Timing:** Verifying period and bit transition accuracy
 - **Sequence Agreement:** Ensuring the two generated sequences aligned correctly.
2. **Synchronization Testing:** A series of tests were designed to assess the system's ability to achieve and maintain synchronization under different conditions. This involved:
 - **Initial Synchronization:** Measuring the time taken to achieve initial lock. Successful synchronization was visually indicated by the illumination of the synchronization LED.
 - **Impact of Data Transmission on Synchronization:** A critical aspect of the DSSS system is its ability to maintain synchronization even while transmitting data. This is achieved through the use of an RS latch in the synchronization circuit. The RS latch ensures that the received signal during data transmission, which cannot be used for direct synchronization due to the data modulation, does not disrupt the established synchronization lock.
3. **Demodulation and Data Recovery:** The balanced demodulator's effectiveness in recovering the original data stream from the received DSSS signal was rigorously evaluated.
4. **Spectrum Analysis:** A spectrum analyzer was used to observe and compare the frequency characteristics of key signals throughout the DSSS system. This included:
 - **Original Data:** The spectral content of the original data signal.
 - **Spreading Signal:** The spectral content of the pseudorandom noise (PN) sequence used for spreading.
 - **BPSK Modulated Signal:** The spectral content of the BPSK modulated DSSS signal.
 - **Received Data:** The spectral content of the recovered data after despreading and demodulation.

6.1. Equipment

The below equipment was used:

- **Oscilloscope:** HAMEG Analog Digital Scope HM407 (40MHz, 100Ms/s) - Utilized for signal visualization and timing measurements throughout the DSSS system.
- **Function Generator:** INSTEK GFG-8217 - Employed as the clock source (0-5V output), providing clock signals to drive the CMOS and TTL ICs.
- **DC Power Supply:** GW Laboratory Dual DC Power Supply (0-15V) - Provided the necessary power rails for the DSSS circuit.
- **Spectrum Analyzer:** INSTEK GSP810 (1000 MHz) - Used for frequency domain analysis of key signals in the system.
- **Coaxial Cables with Crocodile Clip Ends:** Utilized for making connections throughout the breadboard implementation.

A comprehensive testing program was undertaken to validate its functionality, verify performance, and identify potential areas for improvement. Testing was conducted at the Electronics Lab within the EEE Department, with additional use of the Antenna Lab for spectrum analysis.

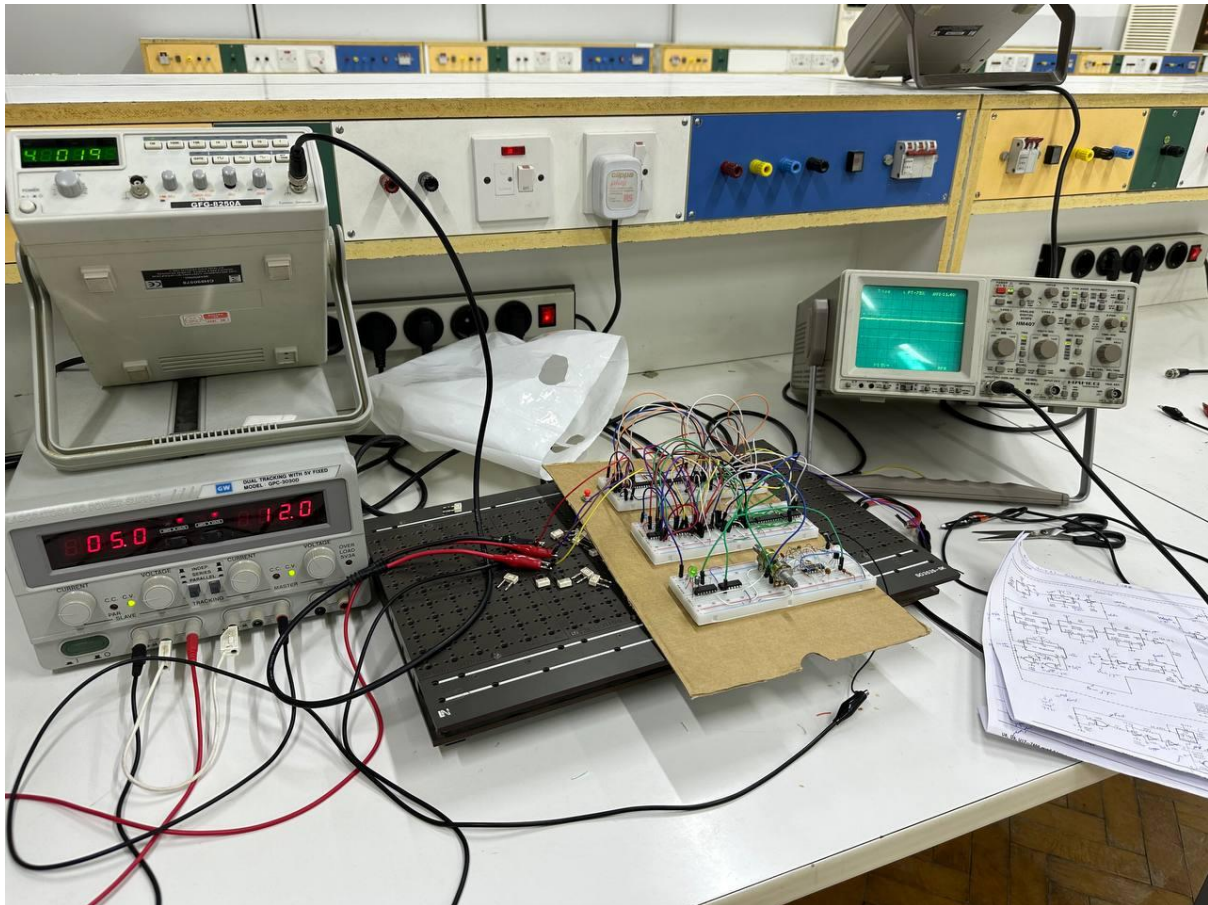


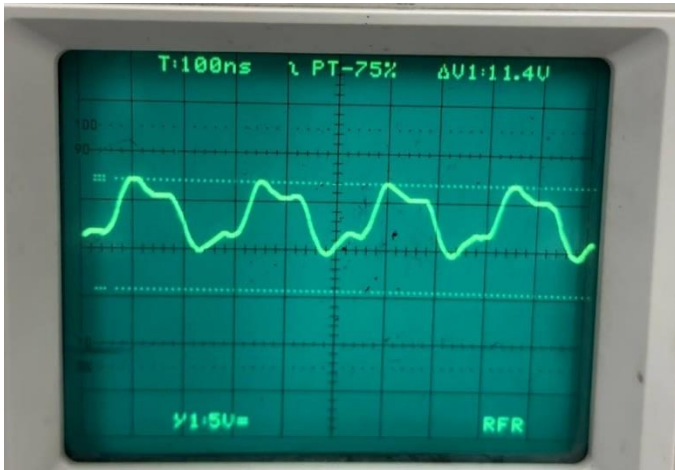
Figure 14. Test Setup at Electronics LAB showing the DSSS circuit connection setup.

6.2. Test Results

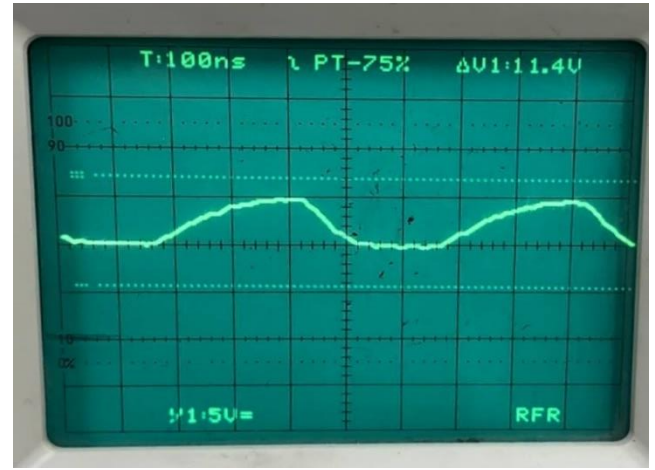
6.2.1. Transmitter

1. Clock Generation and Data Input

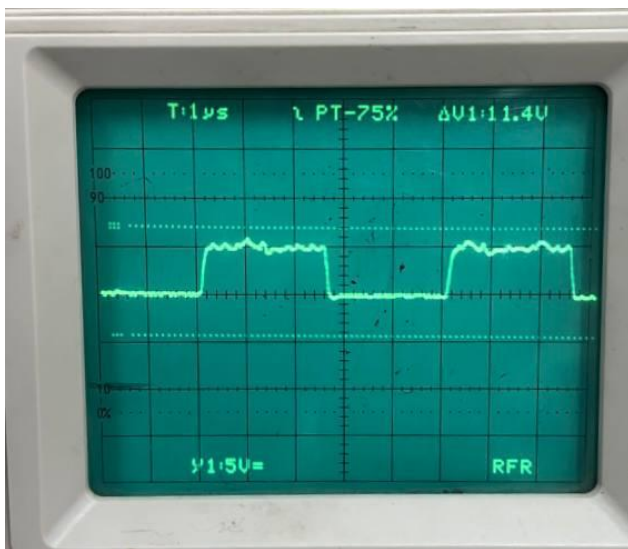
- **Component Overview:**
 - U1: 4-MHz DIP oscillator (reference frequency)
 - U2A: 4013 D flip-flop (divides by 2 for 2-MHz carrier signal, CA)
 - U4, U5, U6: 4017 decade counters (divide-by-10 stages, generating 2-kHz signal)
 - U2B: 4013 D flip-flop (synchronizes Morse code input to 2-kHz clock)
- **Procedure:** The clock generation chain was examined using the oscilloscope, focusing on frequency and transition accuracy. The synchronization of the Morse code input with the clock was also observed.
- **Results:**
 - All frequency division stages were verified, confirming the correct generation of both the 2-MHz carrier and the 2-kHz signal. The Morse code input signal (DA) was successfully synchronized to the 2-kHz clock, as indicated by clean transitions aligned with clock edges.



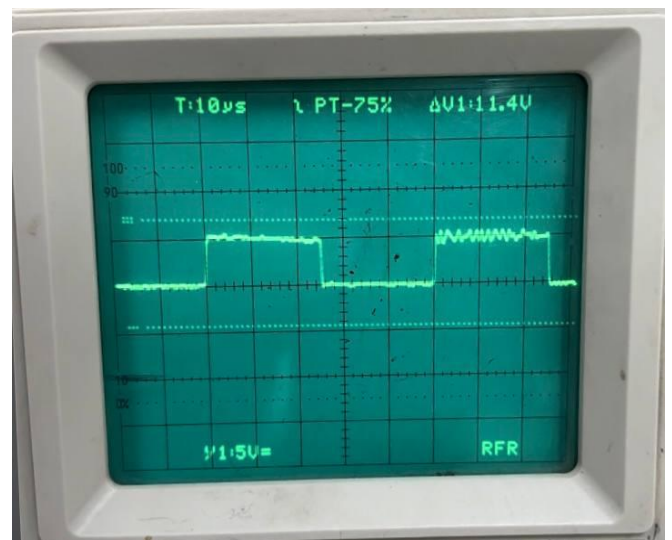
a) 4MHz Reference Carrier from function generator



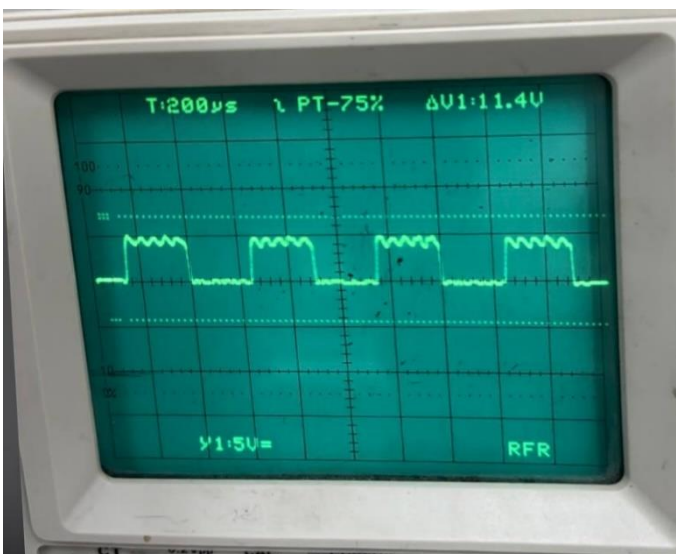
b) 2-MHz carrier signal, CA output from D Flip-flop



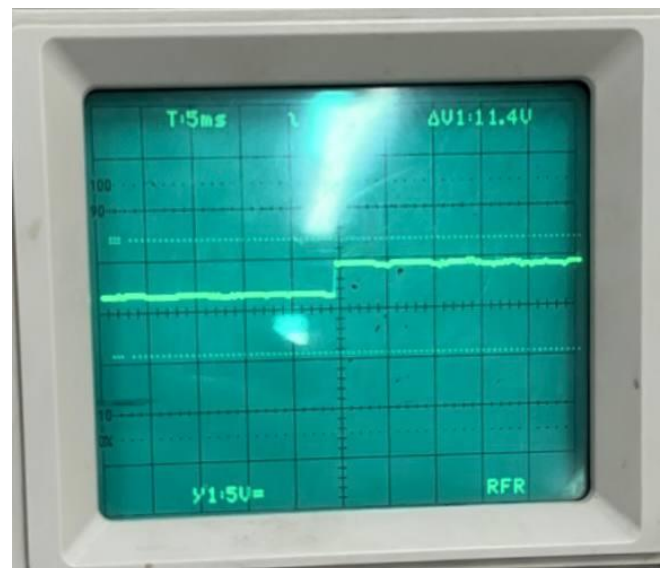
c) 200KHz first Divide by 10 stage



d) 20KHz second divide by 10 stage



e) 2KHz Clock used as Test signal and sync for input data



g) Data input at instant of pressing the push-button

Figure 15. Oscilloscope waveforms for different stages of clock generation and DATA input.

2. PN Sequence Generation

- **Procedure:** The output of the transmitter's PN generator was visualized on the oscilloscope. Timing characteristics and sequence patterns were analyzed.
- **Results:**
 - **Timing:** Both PN generators exhibited the correct bit period and transition timing according to the chosen clock frequency.
 - **Sequence Agreement:** The generated PN sequences were found to be the same as the sequence arrangement of LSFR with 127 bit period, successfully aligning with expectations.

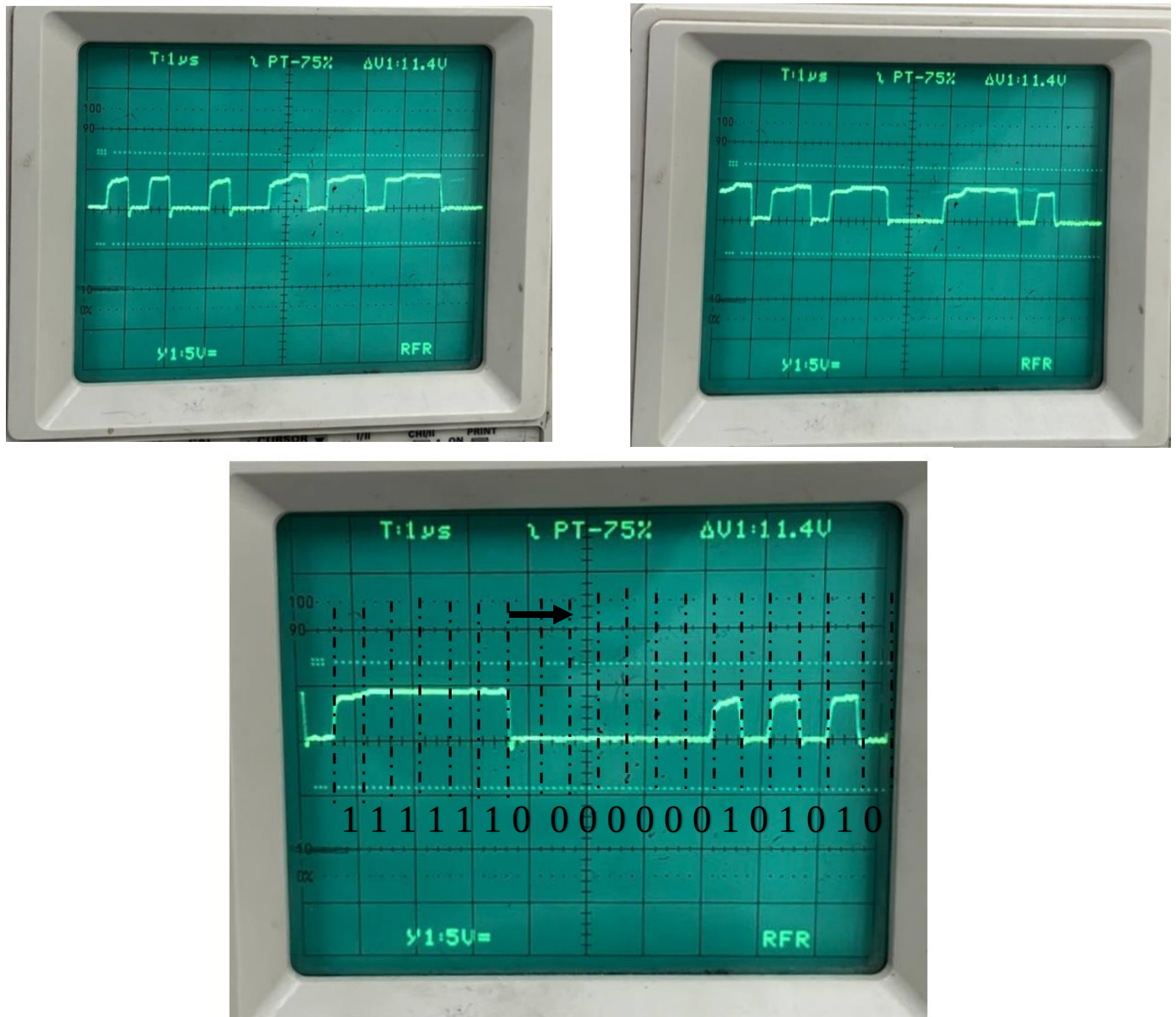
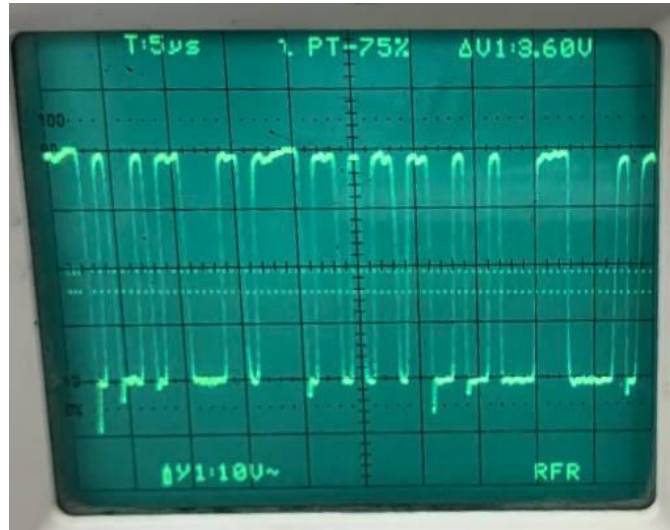


Figure 16. PN generator sequence, the arrow indicates the start of the sequence as appears in Table 1

The bit duration of the PN code as appears $T_c = 0.5 \times 1\mu s = 0.5 \mu s$, which results in a chip rate of $R_c = 1/T_c = 2\text{Mchip/sec}$

- **Spreading**

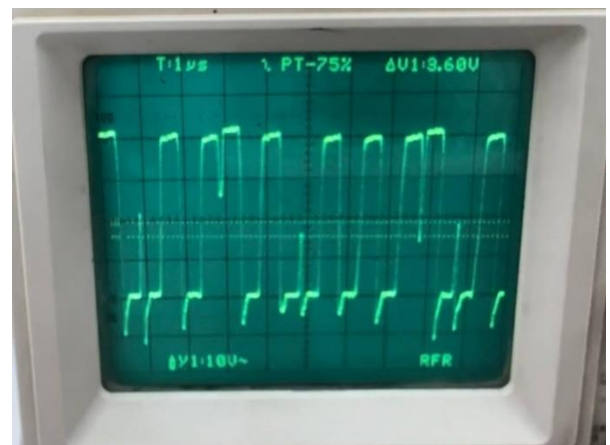
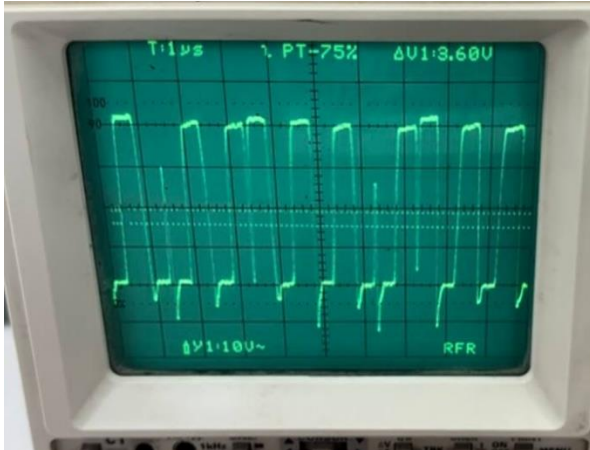
- **Procedure:** The original data and spread data (output of the spreading process) were compared using the oscilloscope.
- **Results:** The oscilloscope confirmed the XOR operation between the data and the PN sequence. This resulted in a spread spectrum signal with a wider bandwidth compared to the original data.
- **Analysis:** The successful XOR operation is a critical step in the DSSS process. It expands the data's spectral occupancy, making it more resistant to interference.



- **BPSK Modulation**

- **Procedure:** The BPSK modulated signal was visualized on the oscilloscope
- **Results:** The oscilloscope confirmed the XOR operation between the spread spectrum data and the carrier signal. This resulted in a BPSK modulated signal with the carrier switching phases (0 and 180 degrees) according to the data bits.
- **Analysis:** The XOR gate effectively implemented BPSK modulation. During a "0" data bit, the carrier remained at its original phase (0 degrees). Conversely, a "1" data bit caused the carrier to invert to 180 degrees. This phase switching encodes the data onto the carrier signal.

Explanation of XOR Gate as BPSK Modulator: An XOR gate acts as a switchable inverter in this context. While the actual BPSK signal may utilize smoother transitions after filtering, the core principle remains. When the spread spectrum data is a "0," the XOR output simply passes the carrier through,



maintaining its 0-degree phase. However, when the data is a "1," the XOR inverts the carrier, resulting in a 180-degree phase shift. This effectively encodes the data onto the carrier for transmission.

6.2.2. Receiver

1. PN Sequence Synchronization

- **Procedure**

- **Initial Synchronization:**

- The transmitter and receiver were started in unsynchronized states.
 - Time between system start and the synchronization LED illuminating was measured using the oscilloscope.

- **Impact of Data Transmission:**

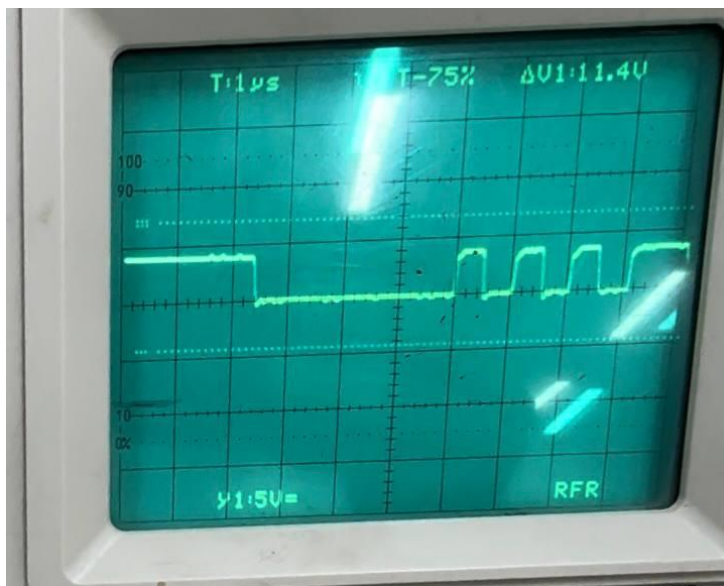
- With established synchronization, data transmission was introduced.
 - Focus was on observing any disruptions to the synchronization LED behavior and potential changes in lock times.

- **Loss of Synchronization:**

- Synchronization was intentionally disrupted (e.g., by momentarily grounding a critical circuit node).
 - The system's ability to regain synchronization and the time taken were observed.

- **Results**

- **Initial Synchronization:** Due to the high 2 MHz clock rate, achieving initial synchronization was effectively instantaneous and could not be reliably measured using the oscilloscope. This rapid synchronization suggests the receiver's PN sequence generator was able to quickly align with the transmitter's sequence within a single clock cycle or a handful of cycles.
 - **Impact of Data Transmission:** As expected, data transmission did not disrupt the established synchronization lock. The synchronization LED remained illuminated, indicating successful data transmission while maintaining synchronization. However, the LED likely exhibited blinking behavior as the magnitude comparator within the synchronization circuit now detected the data modulation within the received signal.



- **Analysis**

- **Meeting Expectations:** The near-instantaneous synchronization observed aligns well with expectations for a system using a high-frequency clock. Faster clock rates provide more opportunities for the receiver's PN sequence to align with the transmitter's during each cycle.

- **Impact of Data Transmission:** The RS latch played a critical role as anticipated. By isolating the synchronization process from the data-modulated received signal, it ensured that the data transmission did not interfere with the established synchronization lock.
- **LED Blinking:** The observed blinking of the synchronization LED is a consequence of the magnitude comparator now detecting the data component within the received spread spectrum signal. While the LED might be visually indicating transitions within the data, the underlying synchronization between the transmitter and receiver's PN sequences remained stable.

2. Demodulation and Data Recovery

• Procedure

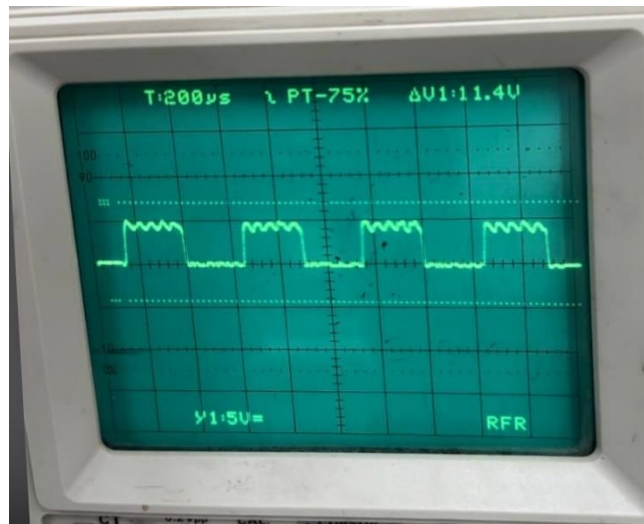
- **Test Signal:** A 2 kHz square wave was used as the input data signal for the DSSS system.
- **Observation Method:** The original 2 kHz square wave and the demodulated output signal were compared on the oscilloscope, focusing on waveform shape and potential differences.

• Results

- **Distortion Under Ideal Conditions:** While the demodulated signal largely resembled the original 2 kHz data, slight distortion was observed within the waveform.

• Analysis

- **Demodulator Performance:** Despite the distortion, the demodulator did successfully recover the fundamental 2 kHz frequency component of the test signal. Which calls for more sharp filtering required.



6.2.3. Spectrum Analysis Findings

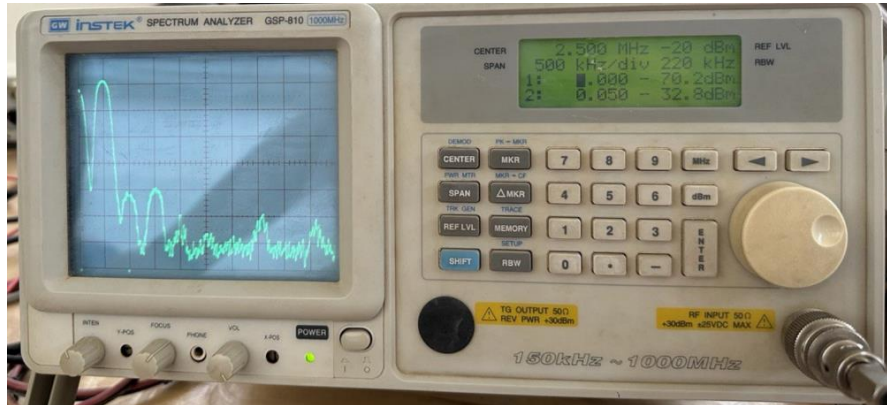
• Procedure:

- The spectrum analyzer was connected to key points in the DSSS system to observe the frequency content of the following signals:
 - Original Data
 - Spreading Signal (PN Sequence)
 - BPSK Modulated Signal
 - Received Data (After Despreading/Demodulation)
- Relevant settings (frequency range, resolution bandwidth) were adjusted based on the expected signal characteristics.

- **Results & Analysis**

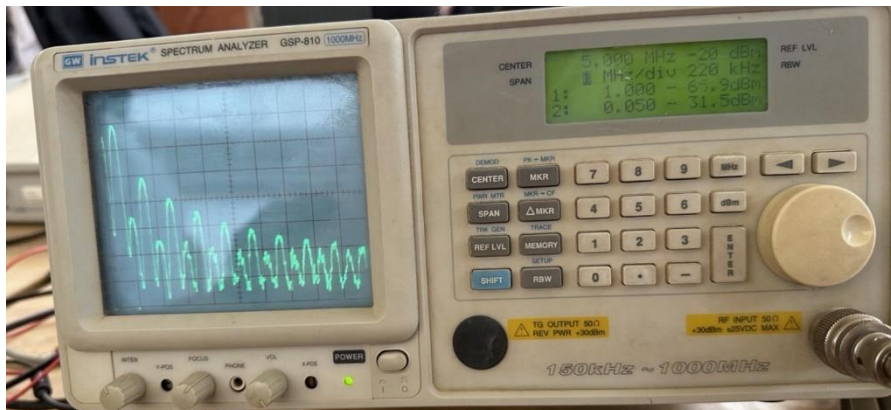
- **Original Data:**

- The spectrum of the original data (e.g., the 2 kHz square wave) should exhibit a relatively narrow bandwidth, with its energy concentrated around the fundamental frequency and potentially some harmonics.



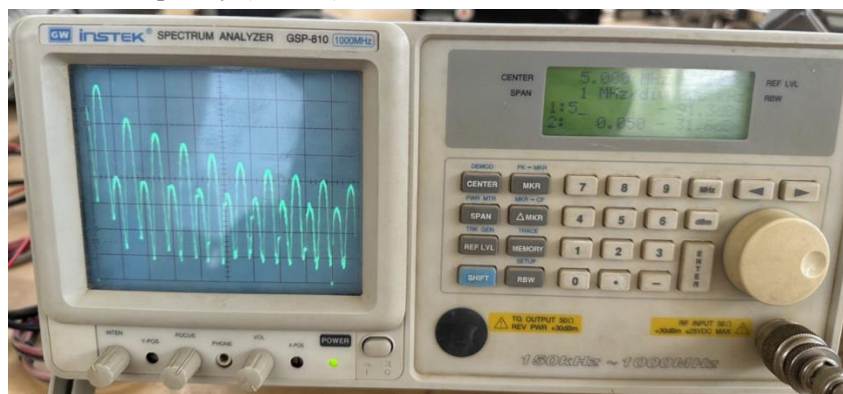
- **Spreading Signal:**

- The PN sequence should have a significantly wider bandwidth compared to the original data. Ideally, it should resemble a noise-like spectrum.



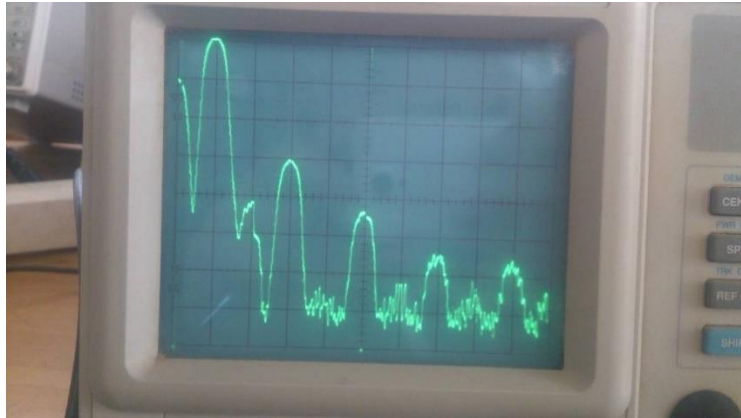
- **BPSK Modulated Signal:**

- The BPSK modulated signal should show spectral spreading similar to the PN sequence. However, the center frequency should now correspond to the carrier frequency (2 MHz).



- **Received Data:**

- After despreading and demodulation, the spectrum of the recovered data should closely resemble the spectrum of the original data signal. This signifies the successful extraction of the original information from the spread spectrum signal.



- **Significance:**
 - Spectrum analysis provides a visual confirmation of the spreading and despreading processes at the core of the DSSS system. Observing the expected spectral changes validates the fundamental operation of the design.

6.3. Process Gain calculation.

Process gain is the improvement in signal-to-noise ratio (SNR) achieved through the DSSS spreading and despreading process. It's a key metric for evaluating DSSS performance.

- **Formula:** The process gain G_p is calculated in decibels (dB) using:

$$G_p = 10 * \log \frac{R_c}{R}$$

where R_c is the spreading bit rate, R is the data rate. In our case $R_c=2MCPS$, $R=2Kbps$

$$G_p = 10 * \log \frac{2 * 10^6}{2 * 10^3} = 30dB$$

The process gain could also be measured in terms of Bandwidth:

Process Gain (dB) = $10 * \log_{10} (\text{Bandwidth of Spread Signal} / \text{Bandwidth of Data Signal})$

Unfortunately, the used spectrum analyzer doesn't have **Occupied Bandwidth (OBW) functionality**, the Bandwidth was measured close to containing approximately 80% of the signal's total power manually:

Bandwidth of Spread Signal= 10.7MHz , Bandwidth of Data Signal= 15.4KHz

→ $G_p = 28.41dB$

The measured process gain of ~30 dB demonstrates the DSSS system's ability to significantly improve the signal-to-noise ratio.

7. PROJECT OUTCOMES AND FUTURE DESIGN EVOLUTION

7.1. Project Summary

This project aimed to design, implement, and thoroughly test a functional Direct Sequence Spread Spectrum (DSSS) communication link. The system successfully demonstrates the core concepts of DSSS, including:

- **Spreading:** The ability to significantly expand the bandwidth of a data signal using a pseudo-random noise (PN) sequence, enhancing its resilience to interference.
- **Synchronization:** The use of a 2 MHz clock enabled near-instantaneous synchronization at the receiver, critical for reliable data recovery in noisy environments. The inclusion of an RS latch ensured stable synchronization even during active data transmission.
- **BPSK Modulation:** Effective integration of Binary Phase Shift Keying (BPSK) for encoding data onto the spread spectrum carrier signal.

These successes highlight the potential of DSSS techniques for secure and robust communication, even in the presence of noise. This project served as a valuable educational tool for understanding the fundamental building blocks of spread spectrum communication. It bridges theory and practical implementation.

7.2. Design Improvements and Optimizations

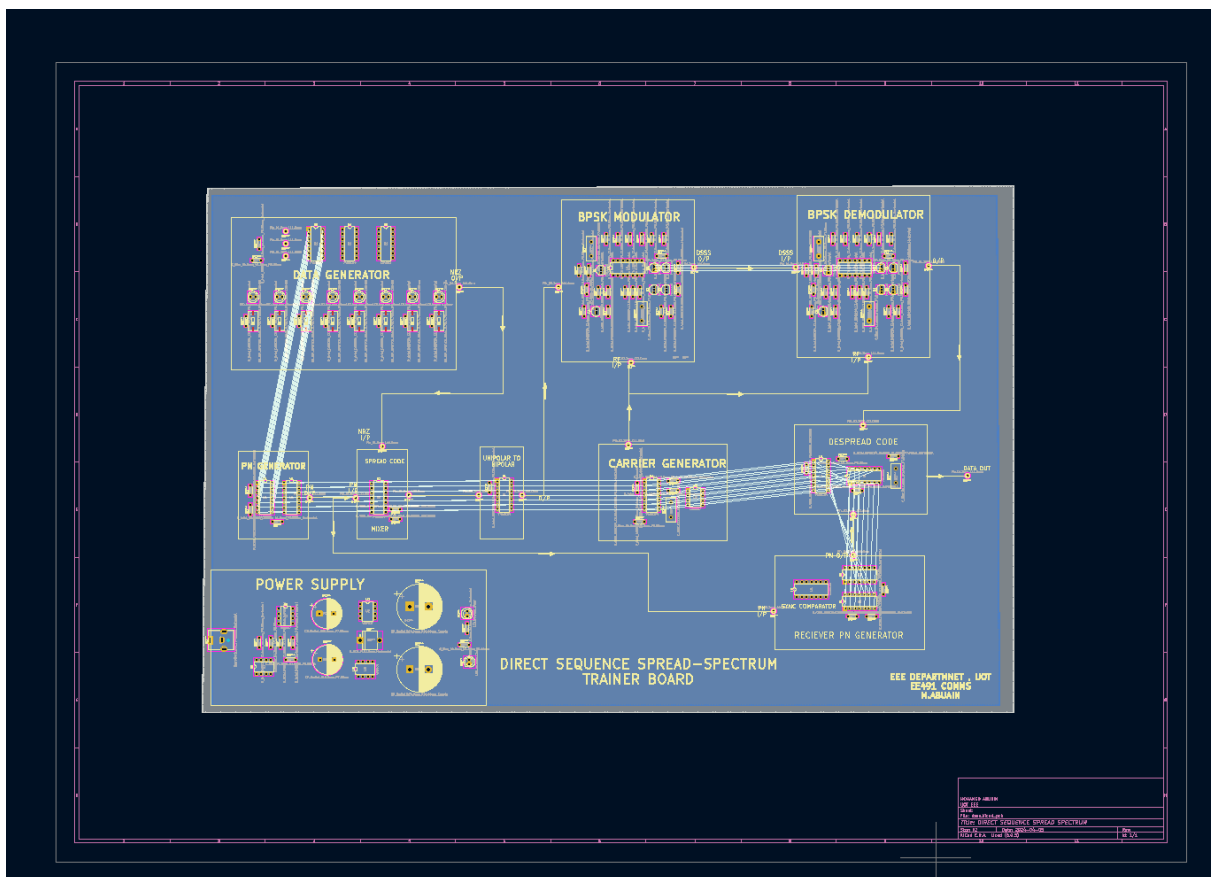
To enhance the performance of the DSSS system, the following design changes could be considered:

- **Faster Components:** Employing faster ICs could improve timing accuracy throughout the system, potentially allowing for more complex modulation schemes or tighter synchronization tolerances.
- **PCB Design:** Transitioning the system from breadboard to a custom PCB would improve overall stability, reduce parasitic effects, and provide a more compact and robust implementation.
- **Adaptation for VHF Operation:** To transition the DSSS system to a VHF radio link, significant changes would be necessary. These include:
 - **Mixer Replacement:** Replacing the XOR-based mixer stages (U8C, U8D, U9A, U17D) with suitable doubly-balanced mixers for VHF operation (component selection would depend on the specific VHF frequency).
 - **Reference Signal:** Utilizing a stable reference carrier within the VHF band or one slightly higher. Implementing a phase-locked loop (PLL) to synchronize a local oscillator to this reference would likely be needed for stability.
 - **Synchronization Preamble:** Consider sending a dedicated preamble signal for synchronization purposes prior to data transmission in the VHF band.
 - **Audio Transmission:** To transmit audio waveforms instead of square waves, increase the base clock rate (e.g., to 20 kHz) and employ a pulse-width modulation scheme using a 555 timer or similar. This setup would facilitate encoding audio signals onto the DSSS carrier.

- **Exploring CDMA Concepts:** A particularly exciting direction for future development would be investigating the adaptation of this DSSS system to the principles of Code Division Multiple Access (CDMA). Key modifications would include:
 - **Unique Spreading Codes:** Assigning each user a unique orthogonal code sequence for spreading their data, allowing multiple users to share the same frequency band at the same time.
 - **Receiver Modifications:** The receiver would require adaptation to be able to distinguish and despread signals from multiple users, potentially needing more complex correlation-based detection.

7.3. Potential Applications

- **Educational Tool:** This DSSS project, particularly with the PCB design (included below), could serve as a valuable trainer module for demonstrating the fundamentals of spread spectrum communication in a hands-on manner. It would be highly relevant for use in communication lab courses such as Wired Communication Systems (EE516) and Wireless Communication Systems (EE526).
- **Beyond Proof-of-Concept:** The transition from a breadboard implementation to a PCB signifies a move beyond a simple proof-of-concept. This makes the DSSS system more robust and suitable for use as a reusable educational tool.
- **PCB Development:** The included PCB is currently a work in progress. With further refinement, it could soon be ready for production, offering a compact and durable platform for DSSS experimentation.



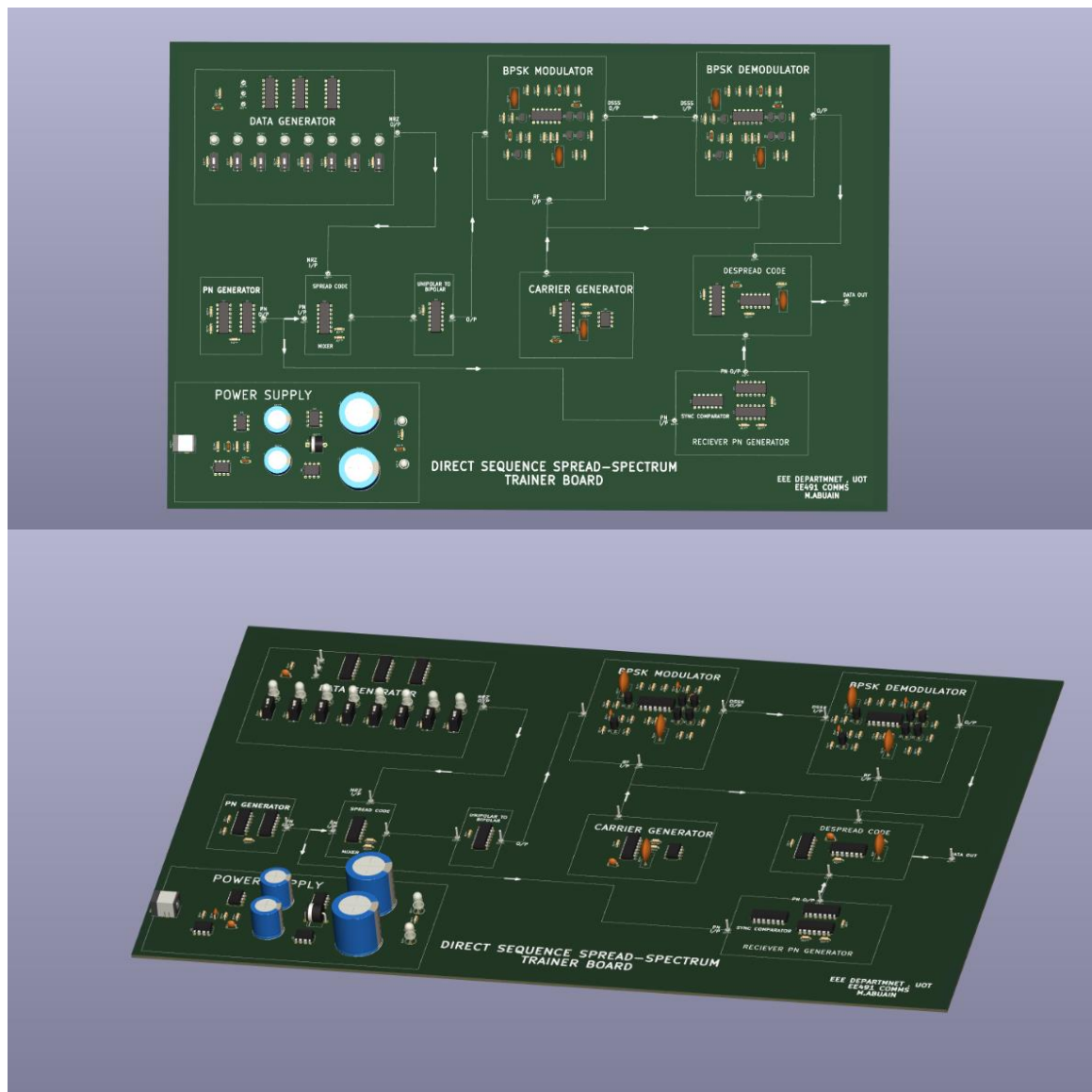


Figure 17. (Work in Progress) PCB Layout (blueprint) along with 3D renders of the PCB. Made with KiCAD.

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